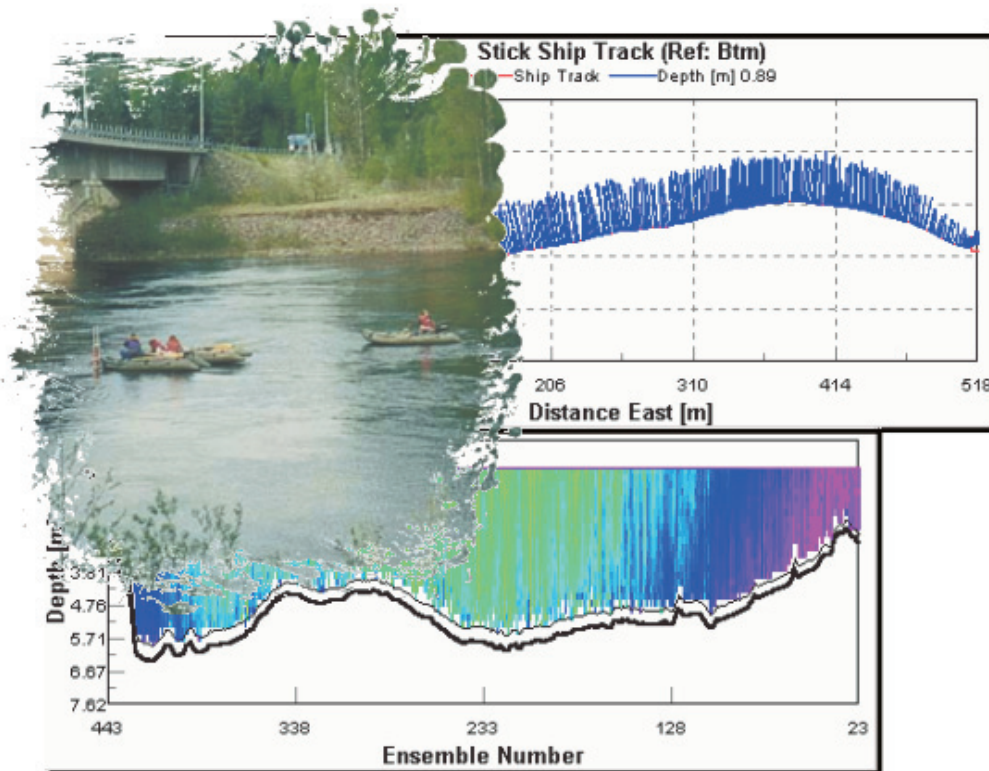


WinRiver II

Quick Start Guide



P/N 957-6230-00 (November 2009)

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WinRiver II Quick Start Guide

Introduction

Thank you for purchasing the Teledyne RD Instruments (TRDI) *WinRiver II* software. This Quick Start Guide will lead you through the steps required for a successful river discharge measurement. Please read the entire guide, and then follow the instructions in the order they are presented. Additional information can be found in the *WinRiver II* User's Guide that is supplied on CD-ROM.



NOTE. To purchase a printed copy of the *WinRiver II* User's Guide, contact our Customer Service department at rdifs@teledyne.com or call +1 (858) 842-2600 and order the *WinRiver II* User's Guide P/N 957-6231-00.



NOTE. This guide covers *WinRiver II* version 2.07.

How to Contact Teledyne RD Instruments

If you have technical issues or questions involving a specific application or deployment with your instrument, contact our Field Service group:

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24 Hour Emergency Support +1 (858) 842-2700

Overview

The first step is to become familiar with the hardware and software that is included with the WorkHorse.

This Section Covers:

- Hardware Overview
- Software Overview
- Installing the Software

Hardware Overview

Take a moment to familiarize yourself with the requirements for a River Discharge measurement.

ADCP Requirements

In order to use *WinRiver II*, your Workhorse ADCP must meet the following criteria.

- WorkHorse Rio Grande ADCP
- or
- The WorkHorse ADCP **must** have the Bottom Track upgrade installed



NOTE. The Bottom Track and High Resolution Water Modes upgrades are available for WorkHorse Monitor and Sentinel ADCPs (included with the Rio Grande ADCP). *This guide assumes you will be using a WorkHorse Rio Grande ADCP.*

The Rio Grande, Monitor / Sentinel, and Mariner ADCPs can benefit from the upgrade to the High Rate Pinging (Water Mode 12) and Shallow Bottom Mode (Bottom Mode 7). Contact your local sales representative if you are interested in upgrading your system.

- *WinRiver II* can also be used with a RiverRay and StreamPro ADCPs if your computer has a Bluetooth connection.



NOTE. In StreamPro firmware version 31.07 the Long Range mode has been extended to 6 meters. For compatibility with WinRiver II v2.06 and above, existing Long Range StreamPro users must have their firmware updated to version 31.07 (or higher).

- *WinRiver II* can also be used with Broadband Phase III systems.

ADCP Mounting Requirements

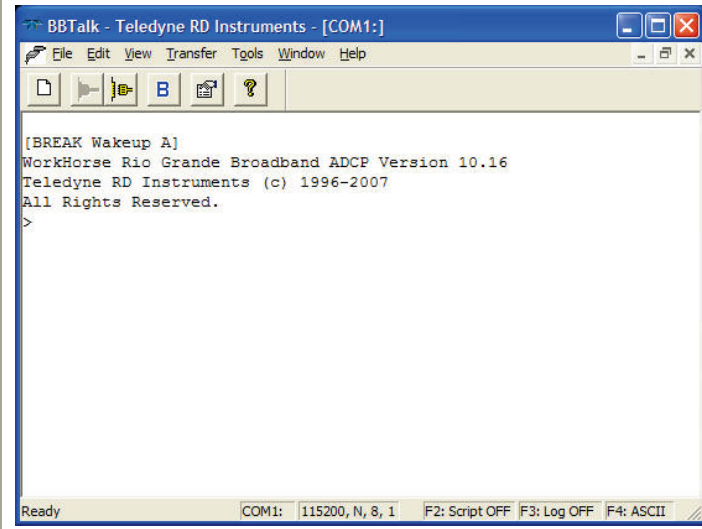
The ADCP must be mounted as follows.

- Over the side or through the hull mounted
- Downward facing
- Within 5 degrees of vertical

TRDI recommends using a fixed mount to achieve the best performance (see the WorkHorse Technical Manual for recommended sources of mounts).

Software Overview

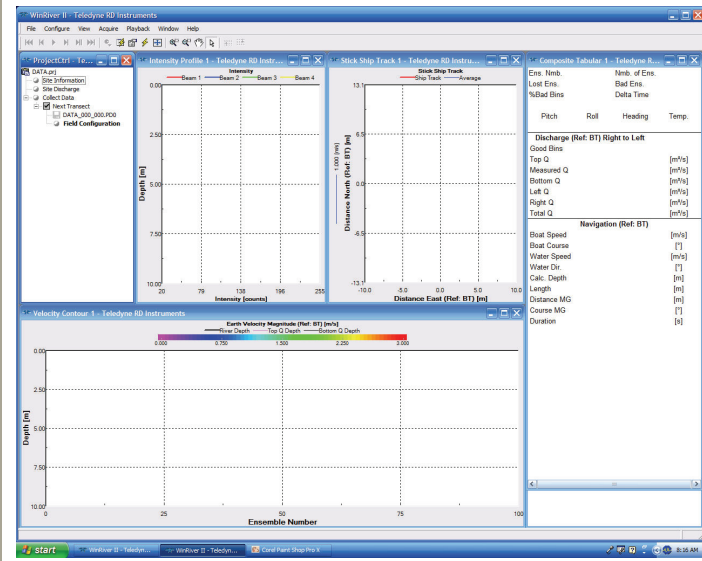
You will use two software programs to test and collect data with the WorkHorse ADCP.



BBTalk Main Screen

BBTalk is a dumb terminal emulator program. It is primarily used to run the ADCP pre-deployment tests and to confirm communications.

This guide does not go into detail about the operation of *BBTalk*. For more information see the RDI Tools User's Guide.



WinRiver II Main Screen

WinRiver II is the software used to set configurations, collect data, and playback previous transects.

Installing the Software

You will be installing two software packages: *RDI Tools* (contains *BBTalk*) and *WinRiver II*. These will be required for testing and measurements.

The software requires the following:

- Windows Vista®, Windows XP Pro® or Windows 2000®
- Pentium III 600 MHz class PC (higher recommended)
- 64 megabytes of RAM (128 MB RAM recommended)
- 50 MB Free Disk Space plus space for data files (A large, fast hard disk is recommended)
- One Serial Port (two or more High Speed UART Serial Ports recommended)
- Minimum display resolution of 1024 x 768
- CD-ROM Drive
- Mouse or other pointing device



Software Installation

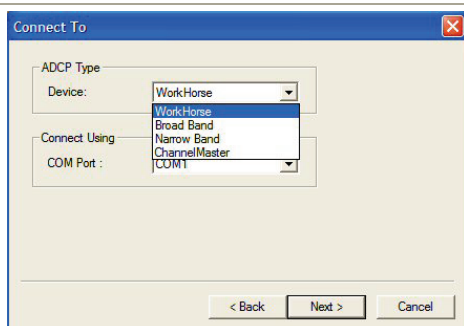
- a. Insert the compact disc into your CD-ROM drive and then follow the browser instructions on your screen. If the browser does not appear, complete Steps "b" through "d."
- b. On the Windows task bar, click the **Start** button, and then click **Run**.
- c. Type <drive>:launch. For example, if your CD-ROM drive is drive D, type d:launch.
- d. Follow the browser instructions on your screen

Setting up Communications

The software needs to have the same configuration as the ADCP / Radio Modems to enable any form of communication.

BBTalk Communications Setup

Use these next steps to confirm communications with the ADCP. See [Commonly Used BBTalk Commands](#) for commonly used commands sent via *BBTalk*. The *BBTalk* software assumes the last communication settings are correct and opens. However, the first time the software is run it is necessary to setup the communications. Therefore, on the first start up, do the following.



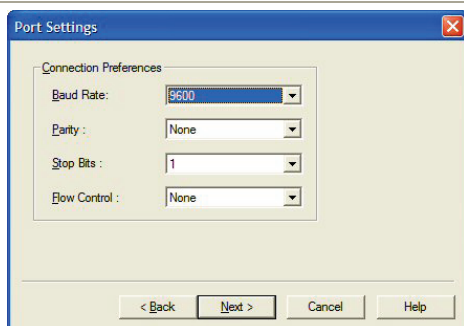
Start BBTalk

Start the *BBTalk* program.

On the **Connect To** screen, select **WorkHorse**.

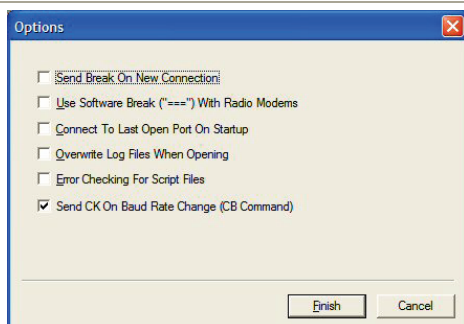
Select the COM port the ADCP cable is connected to.

Click **Next**.



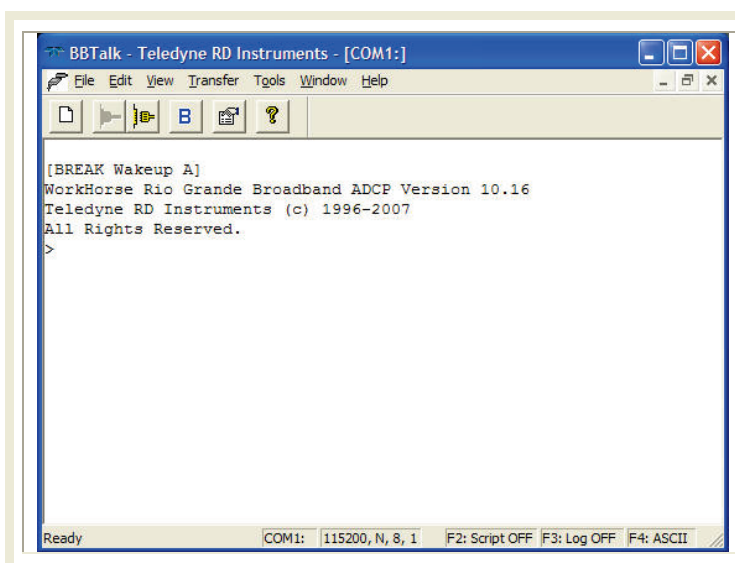
Enter the **Baud Rate**, **Parity**, **Stop Bits**, and **Flow Control**. If you are unsure of the settings, leave them at the default settings.

Click **Next**.



Un-check all boxes except for the **Send CK on Baud Rate Change (CB Command)** box.

Click **Finish**.

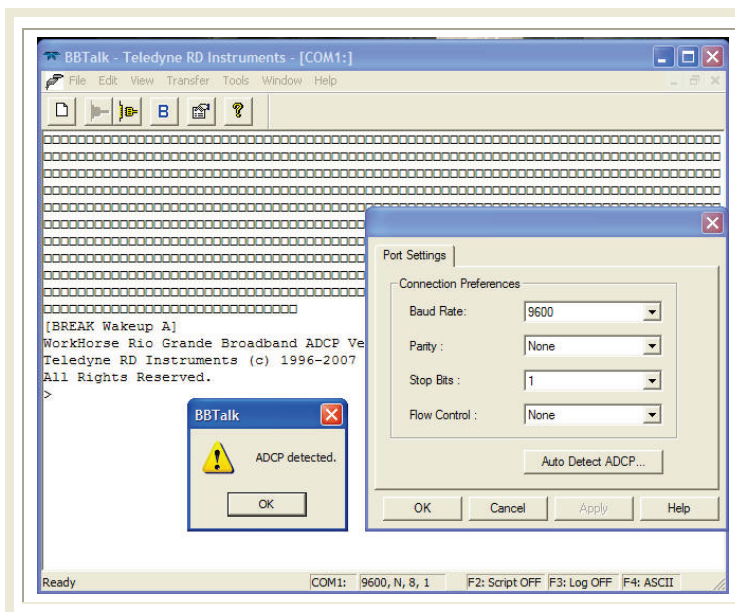


Wakeup

On the **File** menu, click **Break** (you can also press the **End** key to send a break or and press the **B** button on the Tool-bar).

To send a break when using a Radio Modem, press === (3 equals).

You should see the wakeup message appear on the log file window.



If the wakeup message is not readable, do the following.

On the **File** menu, click **Properties**.

Click the **Auto Detect ADCP** button.

Click **OK** when the ADCP is detected. Try to wakeup the ADCP again.

What if the ADCP does not respond?

If your ADCP does not respond, check the serial port, cables, AC power, 12 VDC battery power (Rio Grande only), and battery connection (Sentinel only). If necessary, refer to the Troubleshooting section in the WorkHorse Technical Manual.

Changing the Baud Rate in the ADCPs

The Rio Grande and WorkHorse ADCPs can be set to communicate at baud rates from 300 to 115200. The factory default baud rate is always 9600 baud. The baud rate is controlled via the CB-command. The following procedure explains how to set the baud rate and save it in the ADCP. This procedure assumes that you will be using the program *BBTalk* that is supplied by Teledyne RD Instruments.



NOTE. This applies to WorkHorse ADCPs only. StreamPro and RiverRay ADCPs use 115200 baud rate (see [StreamPro Communication Setup](#)).

```
>
[BREAK Wakeup A]
WorkHorse Rio Grande Broadband ADCP Version 10.15
RD Instruments (c) 1996-2005
All Rights Reserved.
>
>cr1
[Parameters set to FACTORY defaults]
```

Connect the ADCP to the computer and apply power.

Start the *BBTalk* program and establish communications with the ADCP. Wakeup the ADCP by sending a break signal with the **End** key.

At the ">" prompt in the communication window, type **CR1** then press the Enter key. This will set the ADCP to the factory default settings.

BAUD RATE	CB-command
300	CB011
1200	CB111
2400	CB211
4800	CB311
9600	CB411 (Default)
19200	CB511
38400	CB611
57600	CB711
115200	CB811

Send the CB-command that selects the baud rate you wish. The table on the left shows the CB-command settings for different baud rates with no parity and 1 stop bit.

For example, to change the baud rate to 115200, at the ">" prompt in the communication window, type **cb811** then press the Enter key.



The CB? command will identify the communication setting.

```
>cb?
CB = 411 ----- Serial Port Control (Baud
[4=9600]; Par; Stop)
>cb811
>CK
[Parameters saved as USER defaults]
>cb?
CB = 811 ----- Serial Port Control (Baud
[8=115200]; Par; Stop)
>
```

BBTalk will send the command **CK** to save the new baud rate setting.

Exit *BBTalk*.

The ADCP is now set for the new baud rate. The baud rate will stay at this setting until you change it back with the CB-command.

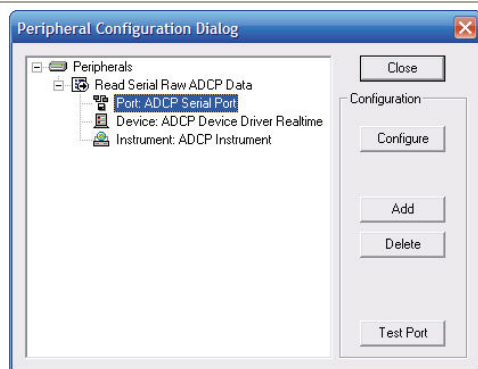


Exit *BBTalk* so the communication port is available for use with *WinRiver II*.

WinRiver II Communications Setup

When *WinRiver II* is first started, you must set up the communications with the ADCP, GPS (if used), external heading (if used), and Depth Sounder (if used). Once setup, *WinRiver II* will remember the settings and use them each time the program is started.

Rio Grande and WorkHorse ADCP Communications Setup



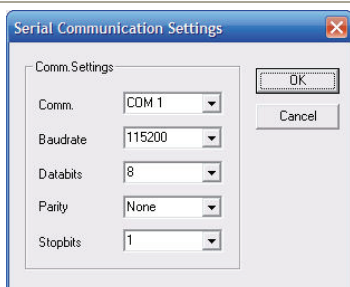
Connect and power up the ADCP as shown in the appropriate ADCP User's Guide.

Start *WinRiver II*.

On the **Configure** menu, select **Peripherals**.

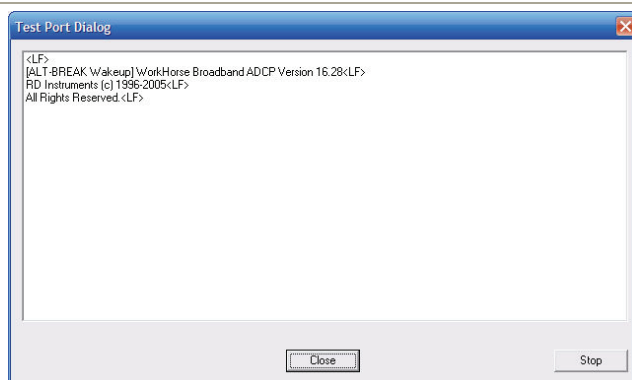
Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Port: ADCP Serial Port**.

Press the **Configure** button.



Select the communication port, Baudrate, Databits, Parity, and Stopbits.

Click **OK** to return to the **Peripheral Configuration Dialog**.



Click the **Test Port** button to connect to the ADCP and confirm the communication setting.

You should see the wakeup message appear on the log file window.

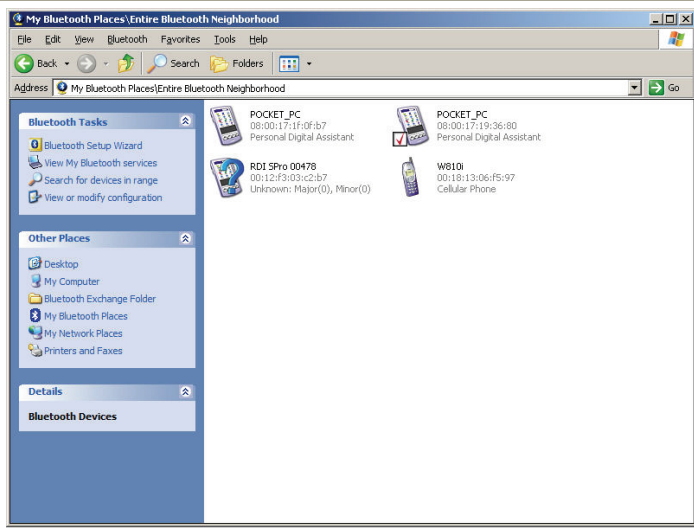
Click the **Close** button to return to the **Peripheral Configuration Dialog**.



If no wakeup message appears, see [Advanced ADCP Communications Configuration](#) and use the **Send Break** button.

RiverRay and StreamPro Communication Setup

WinRiver II can also be used with a RiverRay and StreamPro ADCP if your computer has a Bluetooth connection.



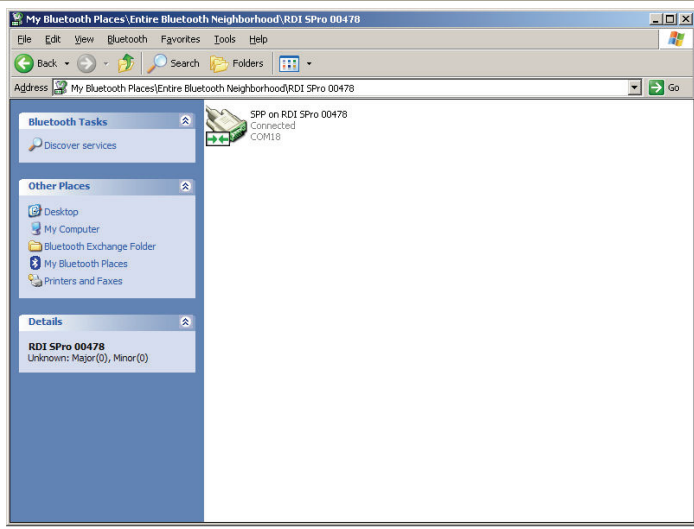
The screenshot shows the 'My Bluetooth Places' window. The address bar indicates 'My Bluetooth Places\Entire Bluetooth Neighborhood'. The main pane lists several Bluetooth devices: POCKET_PC (08:00:17:1F:0F:b7, Personal Digital Assistant), POCKET_PC (08:00:17:19:36:80, Personal Digital Assistant), RDI SPro 00478 (00:12:f3:03:c2:b7, Unknown: Major(0), Minor(0)), and WB10i (00:18:13:06:f5:97, Cellular Phone). The left sidebar shows 'Bluetooth Tasks' with options like 'Bluetooth Setup Wizard', 'View My Bluetooth services', 'Search for devices in range', and 'View or modify configuration'. Below that are 'Other Places' (Desktop, My Computer, Bluetooth Exchange Folder, My Bluetooth Places, My Network Places, Printers and Faxes) and 'Details' (Bluetooth Devices).

Turn on the power to the StreamPro ADCP.

Configure your Bluetooth device. It should locate the RiverRay or StreamPro ADCP (**RDI RRay 00xxx** or **RDI SPro 00xxx** where xxx is the serial number).

 This screen may look different based on what Bluetooth device you are using.

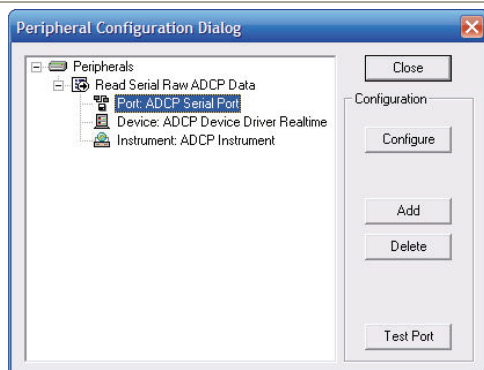
 Some Bluetooth configurations require a pin code for StreamPro and RiverRay ADCPs. The pin code is 0.



The screenshot shows the 'My Bluetooth Places' window with the address bar set to 'My Bluetooth Places\Entire Bluetooth Neighborhood\RDI SPro 00478'. The main pane shows a single device icon for 'RDI SPro 00478' with the status 'Connected COM18'. The left sidebar is the same as the previous screenshot, but the 'Details' pane now shows 'RDI SPro 00478' with 'Unknown: Major(0), Minor(0)'.

Double-click on the **RDI SPro** or **RDI RRay** icon and make sure that that it shows Connected.

Make note of the **Com Port** number it is using. On this example, the StreamPro is connected to Com Port 18.

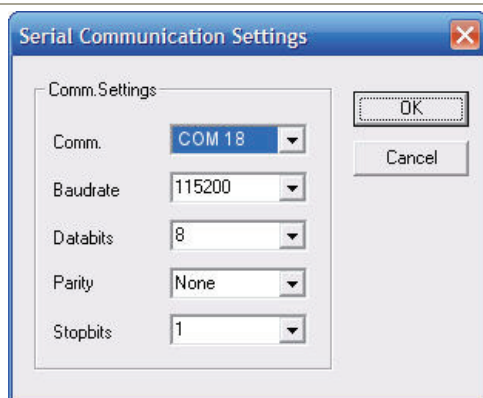


Start *WinRiver II*.

On the **Configure** menu, select **Peripherals**.

Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Port: ADCP Serial Port**.

Press the **Configure** button.

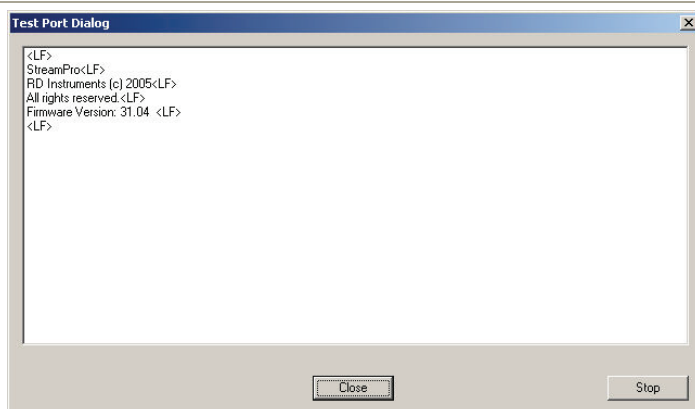


Select the communication port identified when setting up the Bluetooth device (in this example, Com Port 18).

Set the **Baudrate** to 115200.

Leave the **Databits** set to 8, **Parity** to None, and **Stopbits** set to 1.

Click **OK** to return to the **Peripheral Configuration Dialog**.



Click the **Test Port** button to connect to the StreamPro ADCP and confirm the communication setting.

You should see the wakeup message appear on the log file window.

Click the **Close** button to return to the **Peripheral Configuration Dialog**.

Click **Close** once more to exit the communication setting.

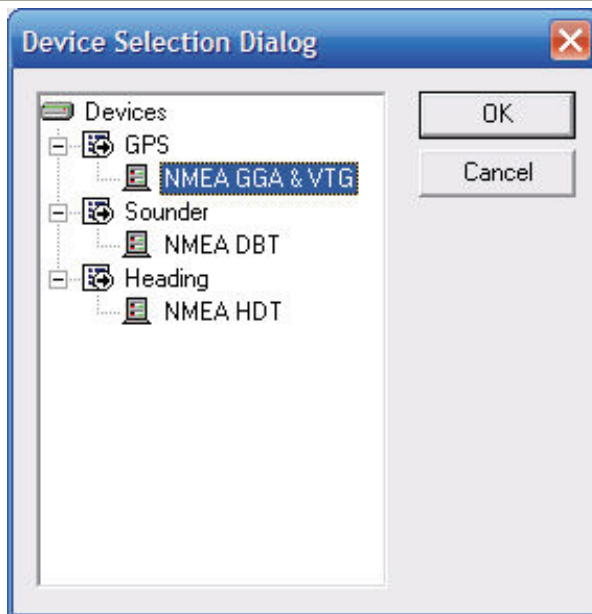


If no wakeup message appears, use *BBTalk* to troubleshoot the communications.



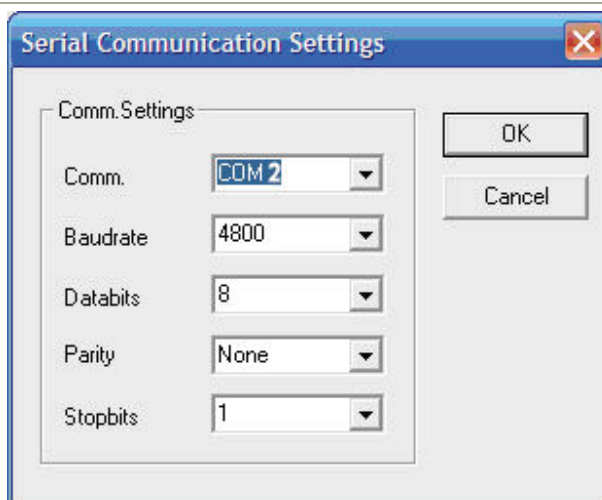
NOTE. The blue LED on the StreamPro ADCP lights when communicating with *WinRiver II*. It may go off when no serial communications are in process. This is normal and does not signify any problem.

Adding GPS, Depth Sounders, or External Heading Devices



On the **Peripheral Configuration Dialog**, press the **Add** button if you will be using GPS, depth sounders, or external heading devices.

Select the NMEA device you want to setup (for example, highlight **NMEA GGA & VTG**) and press **OK**.



Select the communication port, Baudrate, Databits, Parity, and Stopbits and then click **OK** to close the **Serial Communications Settings** dialog.

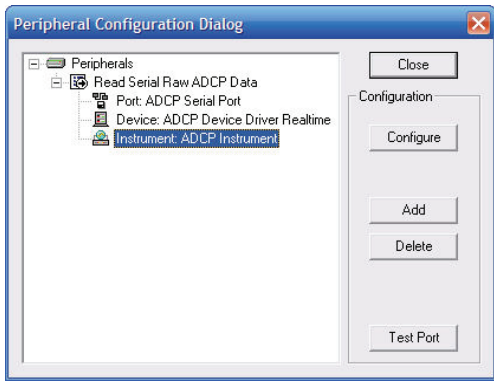
Once all devices have been set up, click **Close** to exit the **Peripheral Configuration Dialog** box.



You will need a serial port for each device you add.

Advanced ADCP Communications Configuration

Use the following steps to communicate with the ADCP or verify command settings.



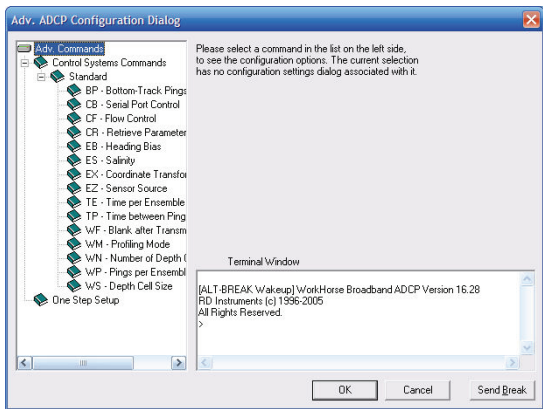
The **Peripheral Configuration Dialog** shows a tree view under **Peripherals** with the following items: **Read Serial Raw ADCP Data**, **Port: ADCP Serial Port**, **Device: ADCP Device Driver Realtime**, and **Instrument: ADCP Instrument**. On the right, there are buttons for **Close**, **Configure**, **Add**, **Delete**, and **Test Port**.

On the **Configure** menu, select **Peripherals**.

Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Instrument: ADCP Instrument**.

Press the **Configure** button.

 Right-click on **Instrument: ADCP Instrument** and select **Auto Detect** to automatically detect the serial port settings.



The **Adv. ADCP Configuration Dialog** has a left pane with **Adv. Commands** and **Control Systems Commands**. Under **Adv. Commands**, there is a list of commands including **BP - Bottom-Track Ping**, **CB - Serial Port Control**, **CF - Flow Control**, **CR - Retrieve Parameter**, **EB - Heading Bias**, **ES - Salinity**, **EX - Coordinate Transform**, **EZ - Sensor Source**, **TE - Time per Ensemble**, **TP - Time between Ping**, **WF - Blank after Transm**, **WM - Profiling Mode**, **WN - Number of Depth I**, **WP - Pings per Ensemble**, and **WS - Depth Cell Size**. A **Terminal Window** is embedded at the bottom, showing the text:
[ALT BREAK Wakeup] WorkHorse Broadband ADCP Version 16.28
RD Instruments (c) 1996-2005
All Rights Reserved.
>
At the bottom of the dialog are **OK**, **Cancel**, and **Send Break** buttons.

Use the **Advanced ADCP Configuration Dialog** screen to send or verify command setting as needed.

If the command you want to send the ADCP is not on the list, type it next to the ">" prompt in the **Terminal Window** and press return.

 Use the **Send Break** button to wake up the ADCP.

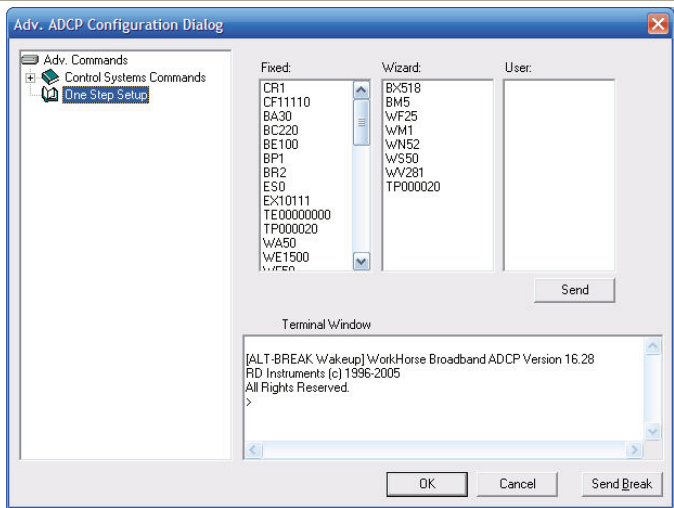


CAUTION. Do not use this screen to deploy or configure the ADCP. The commands will be overwritten when **Acquire, Start Pingging (F4)** is pressed. Only the commands generated through the wizard and sent via **Acquire, Start Pingging (F4)** can be used to deploy the ADCP.



NOTE. WorkHorse Monitor/Sentinel ADCPs will go to sleep if a command is not sent within five minutes; Use the **Send Break** button to wake the ADCP before you start pingging.

WorkHorse Rio Grande ADCPs do not sleep. For more information, see the CL command in the WorkHorse Commands and Output Data Format guide.



Use the **One Step Setup** to check the deployment commands sent to the ADCP after the **Measurement Wizard** has been completed (see [Run the Measurement Wizard](#)).

Run the Measurement Wizard.

Click **One Step Setup**. Click the **Send** button.

If any command generates an error message, stop and correct the problem before deploying.

 This is a good way to check if any command generates an error message.



CAUTION. Do not use this screen to deploy or configure the ADCP. The commands will be overwritten when **Acquire, Start Pinging (F4)** is pressed. Only the commands generated through the wizard and sent via **Acquire, Start Pinging (F4)** can be used to deploy the ADCP.



NOTE. The **One Step Setup** is a good way to check if any command generates an error message (for example, if the commands depend on the High Resolution Water Profiling upgrade but you are not sure the ADCP has this feature).

If a command generates an error, a message box will appear.

The **One Step Setup** does not check the commands for “reasonableness”.

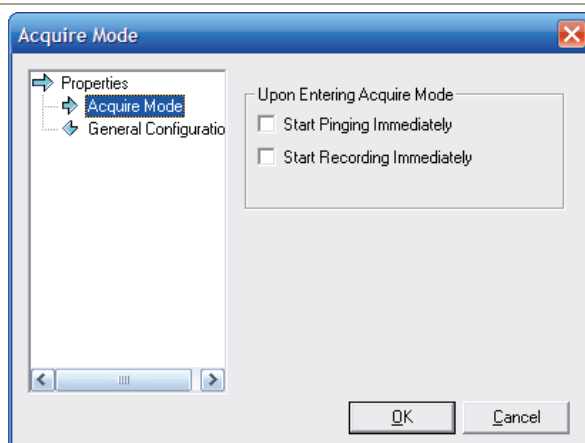
If the wizard has not been run, the **Fixed** and **Wizard** command boxes will be blank.

Commands can be entered in the **User** box or at the “>” prompt.

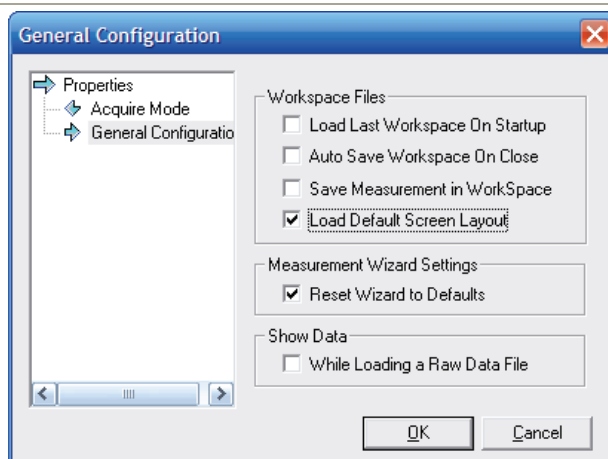
Customizing WinRiver II

WinRiver II can be customized to look and act as you prefer. Once setup, *WinRiver II* will remember the settings and use them each time the program is started.

User Options



On the **File** menu, click **Properties**. The **Properties** dialog sets how *WinRiver II* behaves upon entering the Acquire mode.

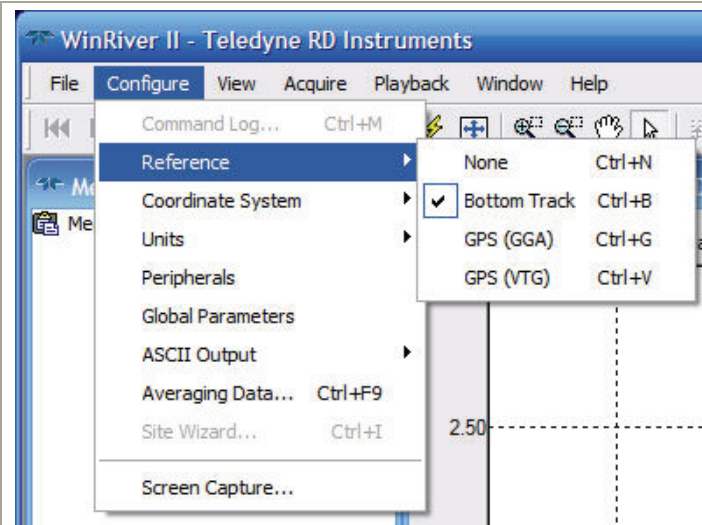


Click on **General Configuration** to change how workspace files are loaded or saved.

Check the **Reset Wizard to Defaults** box to have the measurement wizard use the default settings based on a WorkHorse Rio Grande 600 kHz ADCP.

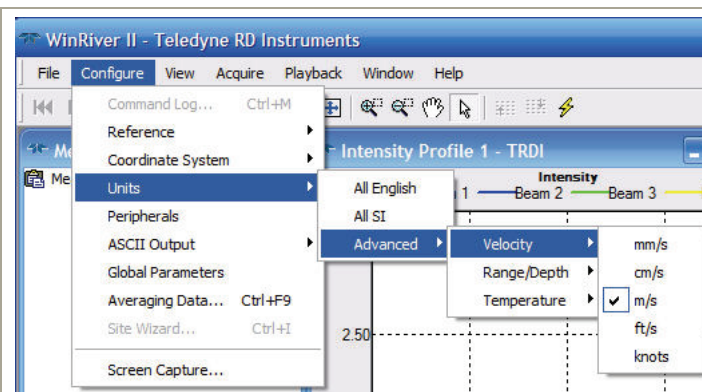
Check the **While Loading a Raw Data File** box to begin displaying data while the file is loading. For smaller data files, this may not be noticeable.

General Preferences



On the **Configure** menu, click **Reference**. Select the desired reference: **None**, **Bottom Track**, **GPS (GGA)**, or **GPS (VTG)**.

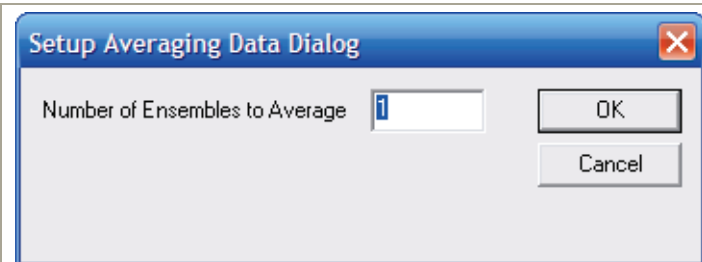
 If the wrong reference is selected during Playback, data may not display. For example, if you select GPS (GGA) as the reference during Playback and this was not collected when the data file was created, no data will display.



To change the units for all displays, on the **Configure** menu, select **Units**.

You can change units to **All English**, **All SI** or use the **Advanced** menu and select each unit for **Velocity**, **Range/Depth**, and **Temperature**.

 When using a StreamPro ADCP on small streams/channels, change the Velocity units to mm/s or cm/sec. This will change the discharge to mm³/s or cm³/s.

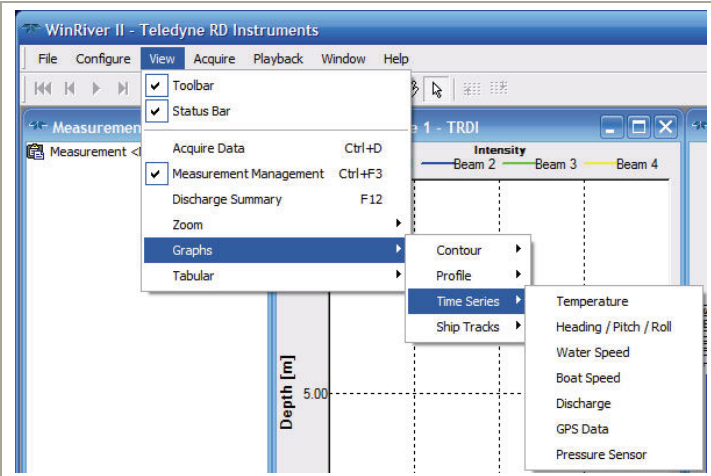


On the **Configure** menu, click **Averaging Data...** Enter a number greater than one to average the data.

 Averaging data may provide smoother plots. This does not alter the raw data in any way.

Creating Workspaces

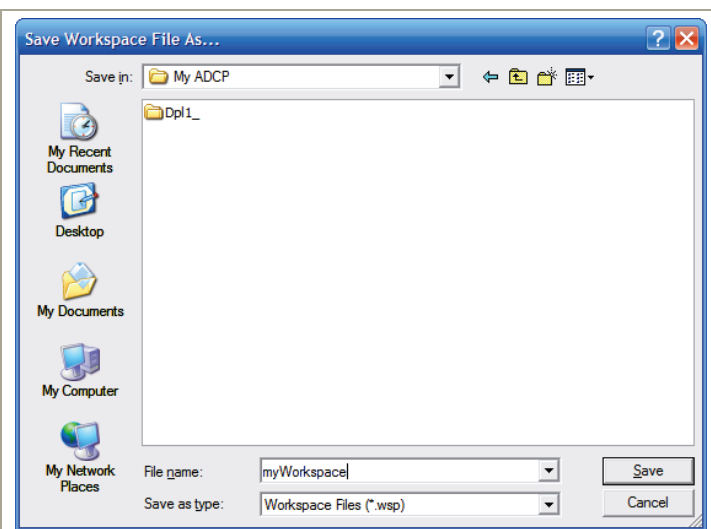
A Workspace is a collection of windows arranged and sized as you prefer, and then saved for future use as needed. It is possible to define as many different workspaces as you would like.



To create a Workspace file, use the **View** menu to open all the windows you want to see during data collection or playback.

Arrange the views you are interested in.

 How workspaces are loaded and saved when starting/closing *WinRiver II* depends on the **User Options** (see [User Options](#)).



When you have the displays set up the way you prefer, on the **File** menu, click **Save Workspace File As**.

To use a workspace, on the **File** menu, click **Load Workspace**.

Collecting River Discharge Data

This section has simple instructions for a typical discharge measurement using the ADCP only (no GPS, External Heading, or Depth Sounder).

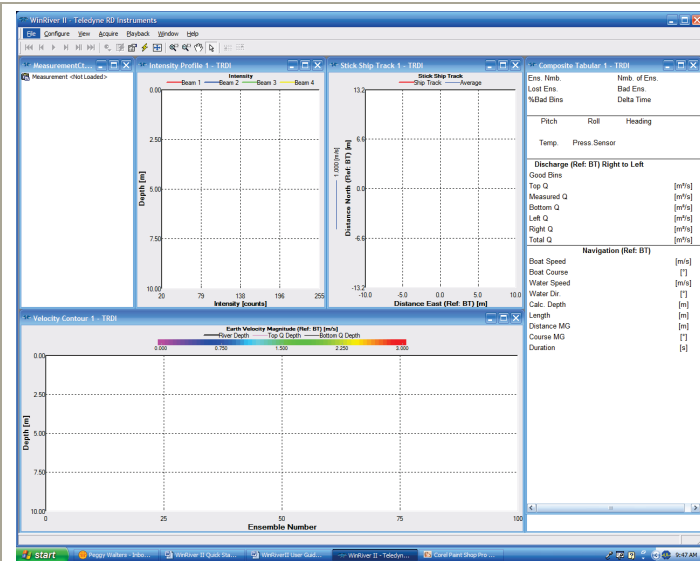
Connect the ADCP



Connect the ADCP and computer as shown in your ADCP User's Guide.

Mount the ADCP on the boat at the desired depth (see the WorkHorse Technical Manual, Installation for details).

Run the Measurement Wizard



Start *WinRiver II*.

On the **File** menu, click **New Measurement** to start the Measurement Wizard.

The Measurement Wizard will create a configuration node and allow you to enter the information needed for taking measurements.



View the tutorials on the WinRiver II software CD for more information.

Site Information

Setup Dialog
Measurement Wizard
Site Information

Station Data

Station Name:

Station Number: Date of Measurement:

River Name: Measurement Number:

Agency Data

Agency:

Country:

State:

County:

District:

Hydrologic Unit:

Field Party Data

Field Party:

Processed by:

Deployment Type:

Boat/Motor:

Meas. Location:

Grid Reference:

Remarks:

Teledyne RD Instruments

Back Next Cancel

Enter the Site Information.

Enter a **Station Name** and **Measurement Number** (alphanumeric). This can be added to the file name (see [Output Filename Options](#)).

Click **Next**.

This information will be included in the Q Measurement Summary (see [Using the WinRiver II Q Measurement Summary](#)).

You can add/edit this information once the measurement wizard is completed by right-clicking on **Site Information** in the Measurement Control window and selecting **Site Wizard**.

Rating Information

Setup Dialog
Measurement Wizard
Rating Information

Rating Information

☒ Inside Gage Height (m): Magn. Variation Method:

☐ Outside Gage Height (m): Measurement Rating:

Gage Height Change (m): Control Code 1:

Rating Discharge (m³/s): Control Code 2:

Index Velocity (m/s): Control Code 3:

Rated Area (m²):

Rating Number:

Water Temp (°C):

Tail Water Level (m):

Teledyne RD Instruments

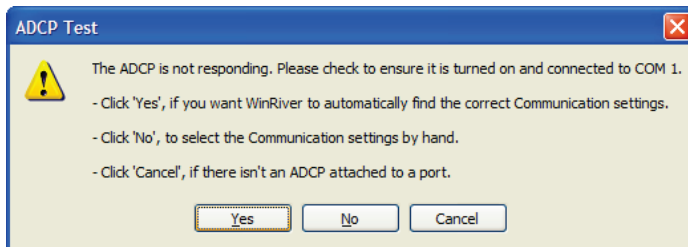
Back Next Cancel

Enter the Rating Information.

Click **Next**.

This information will be included in the Q Measurement Summary (see [Using the WinRiver II Q Measurement Summary](#)).

You can add/edit this information once the measurement wizard is completed by right-clicking on **Site Information** in the Measurement Control window and selecting **Site Wizard**.

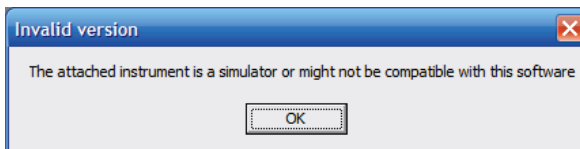


If you see this screen and the ADCP is **NOT** attached to a serial port, click **Cancel** to continue the Measurement Wizard.

If the ADCP is attached to a serial port, click **Yes** to configure the port.



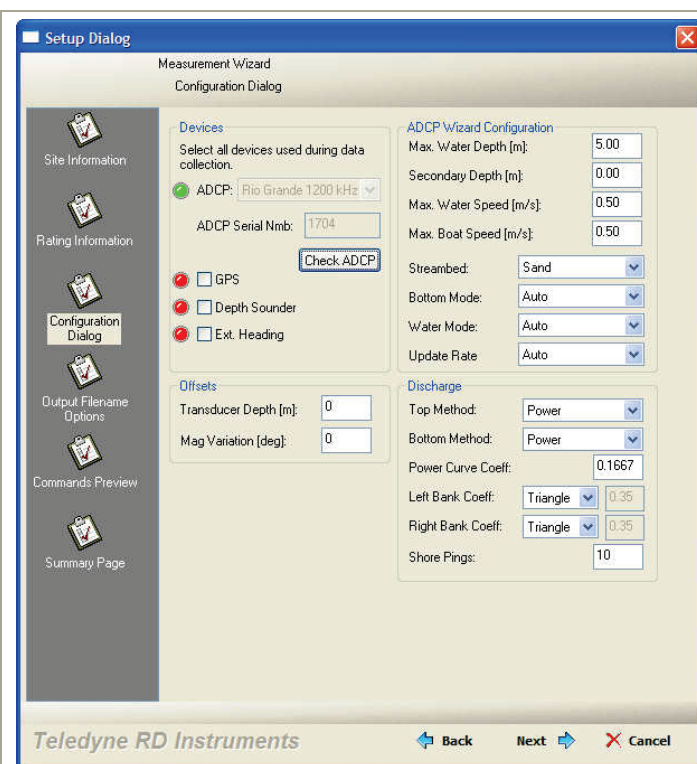
The ADCP does not need to be connected to use the wizard.



If you see the following message box, this means *WinRiver II* is not sure what type ADCP you are using.

Click **OK** to continue.

Configuration Dialog



Enter your choices for how the configuration will be setup.



See the following tables for details.



A green circle means that the communication port has been configured and tested. Red circles mean the device has not been configured.

Devices

Enter your choices for the **Devices** section. See [WinRiver II Communications Setup](#) for instructions on how to setup communications between *WinRiver II* and the ADCP.

If you are using a Rio Grande ADCP, *WinRiver II* will automatically detect and enter the **ADCP Serial Number**, otherwise enter the serial number. Use the **Check ADCP** button to verify the ADCP communications settings.

Selecting the **GPS**, **Depth Sounder**, or **Ext. Heading** boxes will prompt you to set up the communication settings for the device.



The transducer depth should be checked periodically during data collection.

Offsets

Use the **Transducer Depth** field to set the depth from the water surface to the center of the ADCP transducer faces.

Enter the **Magnetic Variation** for the site. See the *WinRiver II* User's Guide for details on how to determine the Magnetic Variation Correction.



It is recommended that the boat speed be less than or equal to the water speed.



Leave the **Bottom** and **Water Mode** set to **Auto**. *WinRiver II* will select a best suited mode based on the water depth, speed, and streambed material.

ADCP Wizard Configuration

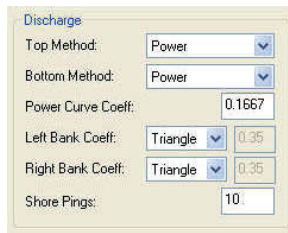
Enter your choices for the **ADCP Configuration** section. Based on the entered information, the wizard will generate the ADCP commands.

The **Max Water Depth**, **Max Water Speed**, and **Max Boat Speed** should be defined as close as possible to the actual conditions.

Set the **Secondary Depth** to the minimum depth that will be measured in the river. Usually this will be a smaller value than the **Max Depth** or left at zero.

Streambed material should then be selected: this does not effect the configuration but it assists in producing relevant warnings.

Bottom Mode and **Water Mode** - See Table 1 for setup table.




For more information on the Discharge settings, see the WinRiver II User's Guide, Discharge Page.

Discharge

Enter your choices for the **Discharge** section.

Select method for calculating the discharge of the unmeasured areas (**Power**, **Constant** or **3pt Slope**). You can also select a coefficient; however, the default **Power** method is standard.

Select the style of the Left and Right banks (**Triangular**, **Square** or **User**). Banks should be selected by facing downstream.

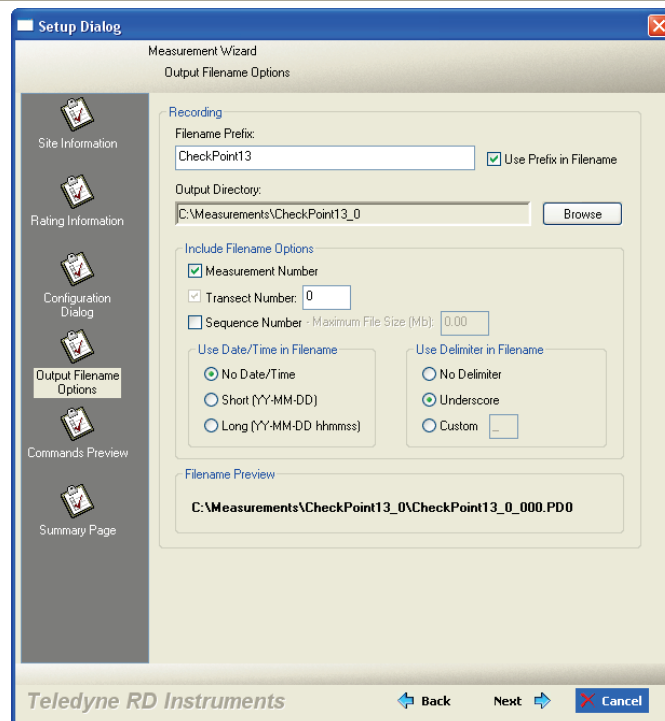
Shore Pings dictates the amount of ensembles that will be used to calculate the edge discharge.

Click **Next**.



You should always use a minimum of 10 shore pings.

Output Filename Options




If a **Station Name** was entered on the first page of the wizard, it will be used in the **Filename Prefix** box.

WinRiver II uses the **Filename Prefix** to create the data file names made during data collection.

Use the **Output Directory** field to select where the data file will be stored.

Check the **Measurement Number** box to add it to the file name (see [Site Information](#)).

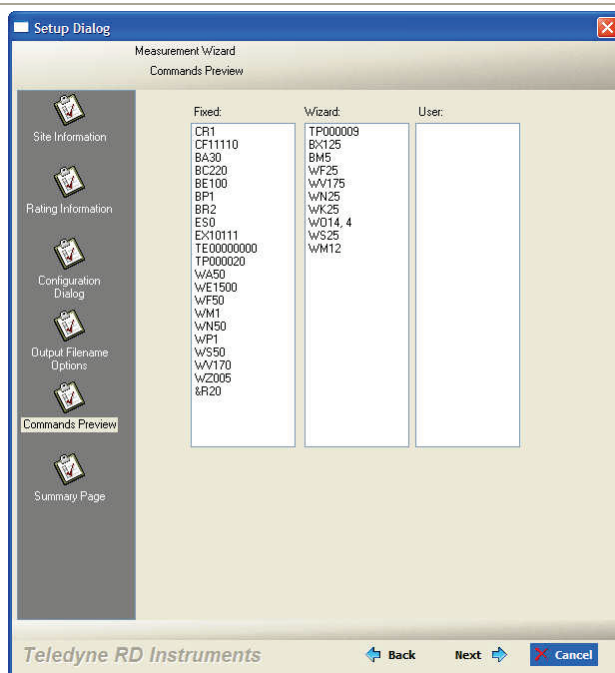
Check the **Sequence Number** box and enter a **Max File Size** if you want to limit the size of the data file. Once the file size has been reached, the sequence number will increment.

To add the Date/Time to the filename, check the **Short (YY-MM-DD)** or **Long (YY-MM-DD hhmmss)** button.

Select what type delimiter to use in the filename by selecting **No Delimiter**, **Underscore**, or **Custom**.

Click **Next**.

Commands Preview



See Table 2 for a short list of ADCP commands commonly sent via *WinRiver II*.

The Commands Preview is where adjustments can be made to the commands produced by the measurement wizard.

There are three columns - the first is the **Fixed** commands that are sent by default by *WinRiver II* to the ADCP.

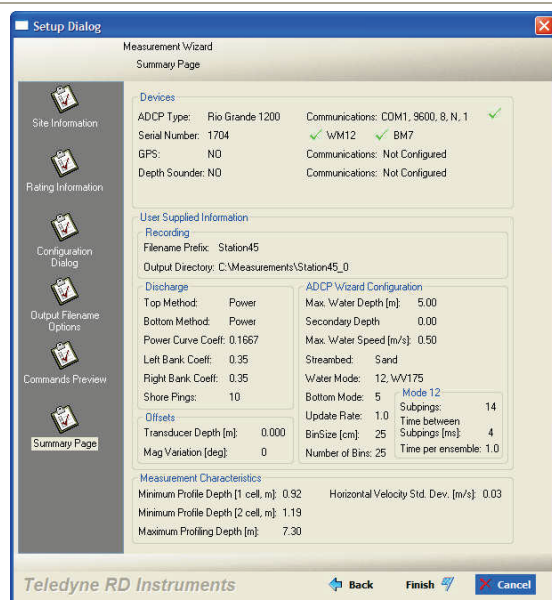
The second column is commands that are produced by the **Wizard**; these commands will overwrite the corresponding original fixed commands.

The final column is for **User** commands; these can be added to by the user, and just as before, these commands will overwrite the previous column.

Do not enter any commands in the User section unless you are fully aware of what the command does.

Click **Next**.

Summary Page



Review the **Summary Page**.

When done, click **Finish**.



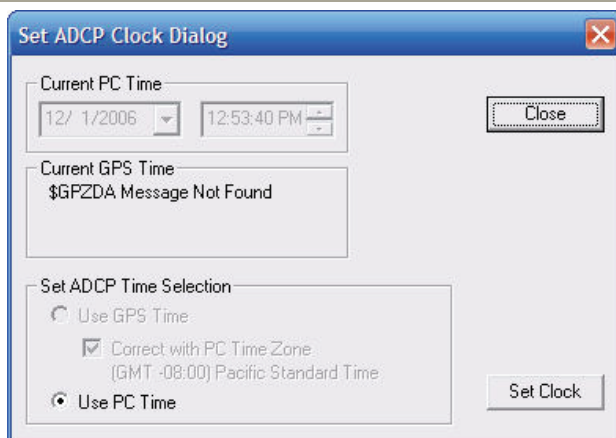
A green check mark next to **WM12** and **BM7** means that the ADCP is capable of using these modes, not that the mode is selected.

View the **ADCP Wizard Configuration** section to see what water and bottom mode is selected.

QA/QC Items

Before taking measurements, check the following items.

Set ADCP Clock



On the **Acquire** menu, click **Set ADCP Clock**.

Click the **Set Clock** button to set the ADCP's time to the GPS time (if available) or the PC's time. If necessary, set the PC's clock first.

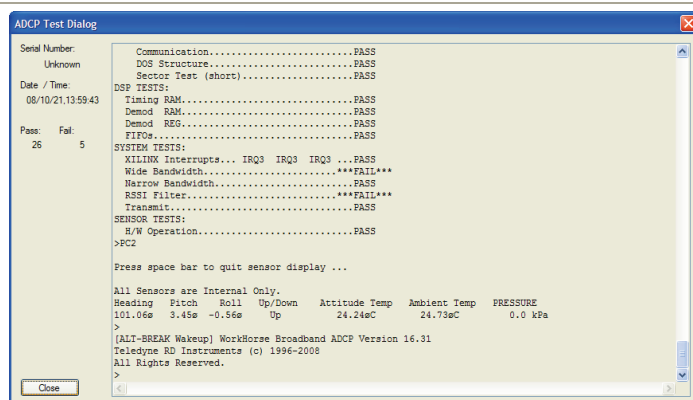


The first time a transect is started, the **Set ADCP Clock** dialog will open.



StreamPro ADCPs users will notice that the blue LED will flash until the **Close** button is clicked. This is normal. The TS command is send once per second until the close button is clicked.

Test ADCP



On the **Acquire** menu, click **Execute ADCP Test** to verify the ADCP is functioning properly.

Click the **Stop PC2** button to end the PC2 test. Click **Close** to exit the **RGTest** dialog.

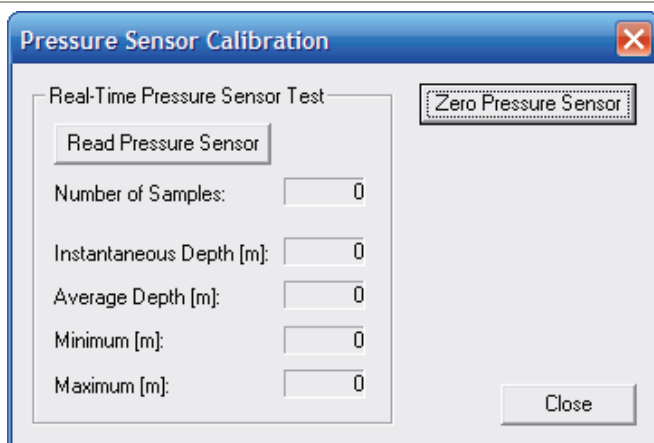


The tests should be run while the ADCP is in non moving water. Running the test in air will not harm the ADCP, but some tests may fail in air.



Do not use the **RGTest** on a StreamPro ADCP. Use **BBTalk** to test a StreamPro ADCP.

Test Pressure Sensor



On the **Acquire** menu, click **Execute Pressure Sensor Test** to verify the ADCP's pressure sensor is functioning properly.

Click the **Read Pressure Sensor** button to get samples.

Click the **Zero Pressure Sensor** button to zero out the sensor.

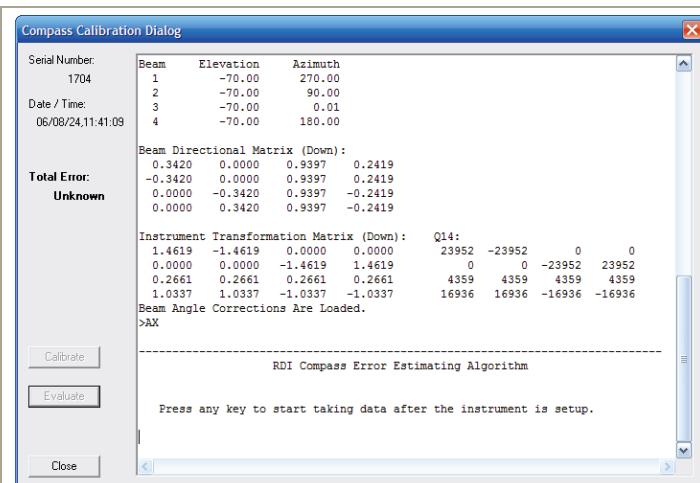


If your ADCP does not have a pressure sensor, you will see an error message.



Do not use the **Pressure Sensor Test** on a StreamPro ADCP. StreamPro systems do not have a pressure sensor.

Compass Calibration



For more information on the Compass Calibration, see the WinRiver II User's Guide, Integrating Depth Sounder, External Heading, and GPS Data.

On the **Acquire** menu, click **Execute Compass Calibration**. Click on the **Calibrate** button to begin the compass calibration.



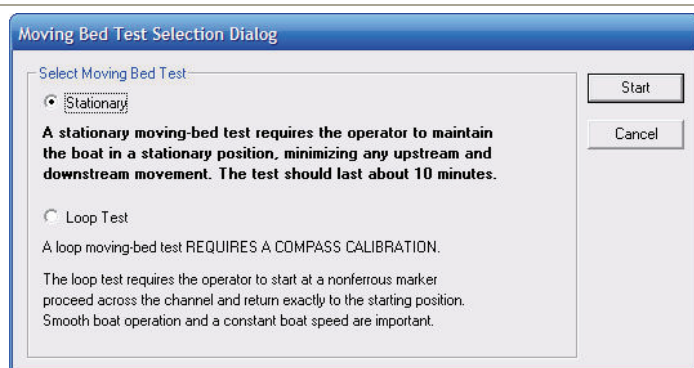
If you will be using GPS rather than Bottom Track as the reference, then the compass must be calibrated and the Magnetic Variation for the site entered on the Configuration Dialog page (see [Configuration Dialog](#)).

See the *WinRiver II User's Guide* for details on compass calibration and magnetic variation corrections procedures.

Moving Bed Test

During high flow season or where the river sediment load is high, acoustic absorption and scattering interfere with the bottom tracking of ADCPs. The Moving Bed test should be performed at every site gauged, and made every time the site is visited as conditions do change. The test can also act as a pre-survey; the data retrieved can be used in the Measurement Wizard to set the depth and velocity.

The Moving Bed test is to prove that the bed of the section is not in motion. If you obtain biased bottom track data at your river site, moving to a new section may help, but flood conditions may require the use of GPS.



The **Loop Test** method was developed by USGS and is not directly supported by TRDI. For more information, refer to the USGS document concerning the loop method at the following link:
<http://pubs.usgs.gov/sir/2006/5079/>

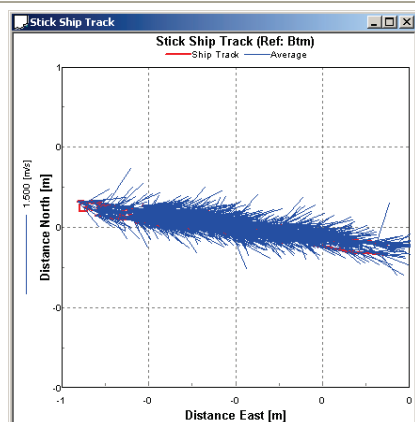
Using the **Measurement Wizard**, set an estimated depth and velocity, and use **Water Mode 1** and **Bottom Mode 5**. These modes can be varied to suit the section better.

On the **Acquire** menu click **Start Pinging** or use the shortcut key **F4**.

Move the ADCP to the middle of the section, or the point at which the highest velocities can be seen.

On the **Acquire** menu click **Moving Bed Test**. Select **Stationary** and click the **Start** button.

Hold the unit in position for ten minutes and try to minimize any movement.



Ship track indicating NO Moving Bed.
Graph scales indicate minimum movement.

Keep a close eye on the **Ship Track Plot**. Any movement indicated on here above actual movement would indicate a Moving Bed.

On the **Acquire** menu click **Stop Moving Bed Test**.

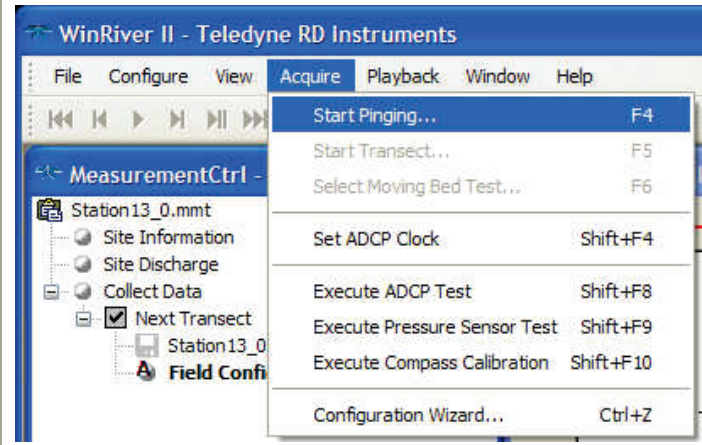
Review the data; if this indicates bed movement, move to a more suitable section.

Save the measurement project.

Use data to set a better configuration for transects.

Mark Transect Start and End Points

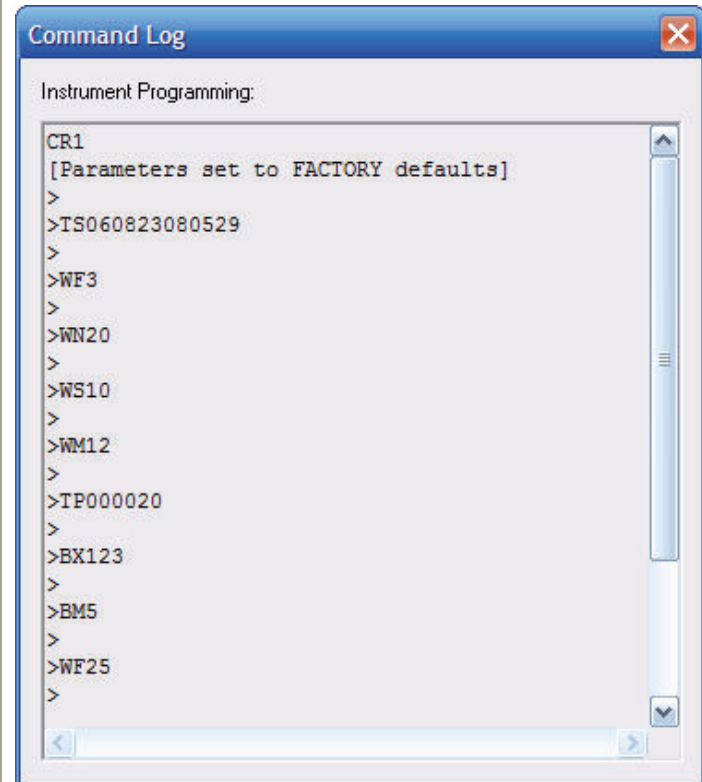
You must determine the start and end positions prior to beginning measurements as *WinRiver II* uses this data to extrapolate for the edges. Typically, 10 shore ensembles are taken as close to the riverbank as can be measured and still read valid data. Ensembles that contain a minimum of two good depth cells are considered valid data. Once these points have been determined, they should be marked and used for the remaining transects.



Start Pinging

On the **Acquire** menu click **Start Pinging** or use the shortcut key **F4**.

 If this is the first measurement taken, *WinRiver II* will prompt for the clock to be set (see [Set ADCP Clock](#)).



WinRiver II will send the commands generated by the configuration wizard to the ADCP.

Review the **Command Log** to see if any error occurs.

 If the command log closed before you finished reviewing the commands, click **View, Command Log...** Scroll through the log file and make sure no command generated an error message.

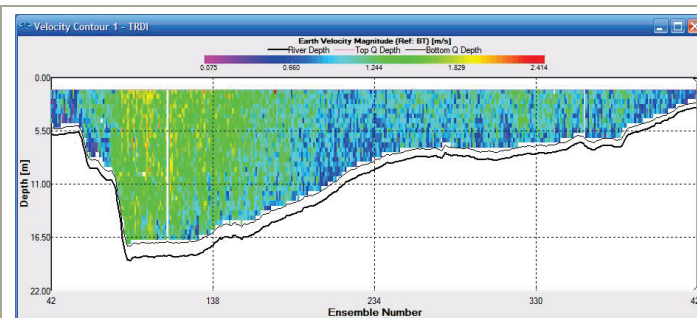
Velocity Tabular 1 - TRDI

Depth [m]	Velocity [m/s] (Ref: BT)				%	Discharge [m³/s]
	East	North	Up	Error		
1.46	-0.230	0.645	0.129	-0.275	100	0.045
1.96	-0.282	0.244	0.138	-0.244	100	0.018
2.46	-0.168	0.072	0.225	0.672	100	0.006
2.96	-0.290	0.818	0.329	0.336	100	0.057
3.46	-0.288	0.536	0.246	-0.055	100	0.038
3.96	0.114	0.338	0.095	0.156	100	0.022
4.46	0.545	0.491	-0.060	-0.635	100	0.030
4.96	BAD	BAD	BAD	BAD	BAD	BAD
5.46	BAD	BAD	BAD	BAD	BAD	BAD
5.96	BAD	BAD	BAD	BAD	BAD	BAD

Click **View, Tabular, Velocity** to open a **Velocity Tabular 1** display.

Move out from the shore until the water is deep enough to consistently show good values for two bins (or more depending on river conditions and how close you can get to the shore). Values of 0.0 are acceptable, but “bad” values are invalid.

Mark this position (with a float). This is the starting/stopping position for this shore. You will later start/stop data file recording at this location depending on the direction of your transect.



Move out from the shore traveling slowly with the bow of the boat pointed up-stream.

💡 Use the **Velocity Magnitude Contour** display to see how the water depth changes as you make your transect. Note regions where the bottom depth changes quickly.

Velocity Tabular 1 - TRDI

Depth [m]	Velocity [m/s] (Ref: BT)				%	Discharge [m³/s]
	East	North	Up	Error		
1.46	-0.490	0.617	-0.023	0.248	100	0.011
1.96	-0.126	0.274	-0.011	0.287	100	0.005
2.46	BAD	BAD	BAD	BAD	BAD	BAD
2.96	BAD	BAD	BAD	BAD	BAD	BAD
3.46	BAD	BAD	BAD	BAD	BAD	BAD
3.96	BAD	BAD	BAD	BAD	BAD	BAD
4.46	BAD	BAD	BAD	BAD	BAD	BAD
4.96	BAD	BAD	BAD	BAD	BAD	BAD
5.46	BAD	BAD	BAD	BAD	BAD	BAD
5.96	BAD	BAD	BAD	BAD	BAD	BAD

Switch back to the **Velocity Tabular 1** display. When you approach the other shore, mark the closest distance to shore where two depth cells show discharge values.

This will be the start/stopping point for this shore.

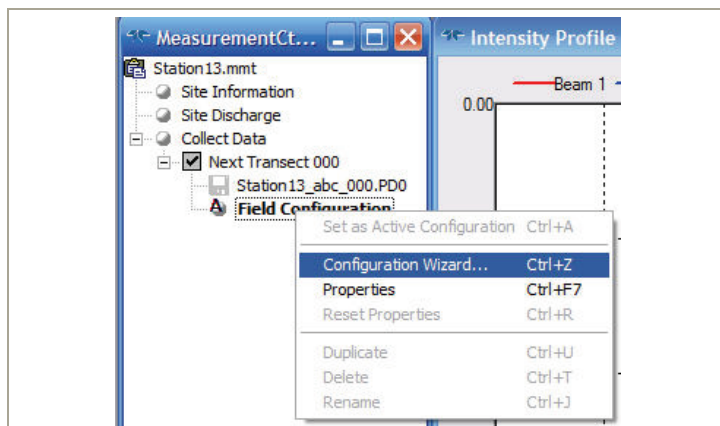
💡 Start/Stop distances from the center of the transducer to the shore should be measured as accurately as possible - at least to the nearest deci-meter.

Adjust the Configuration

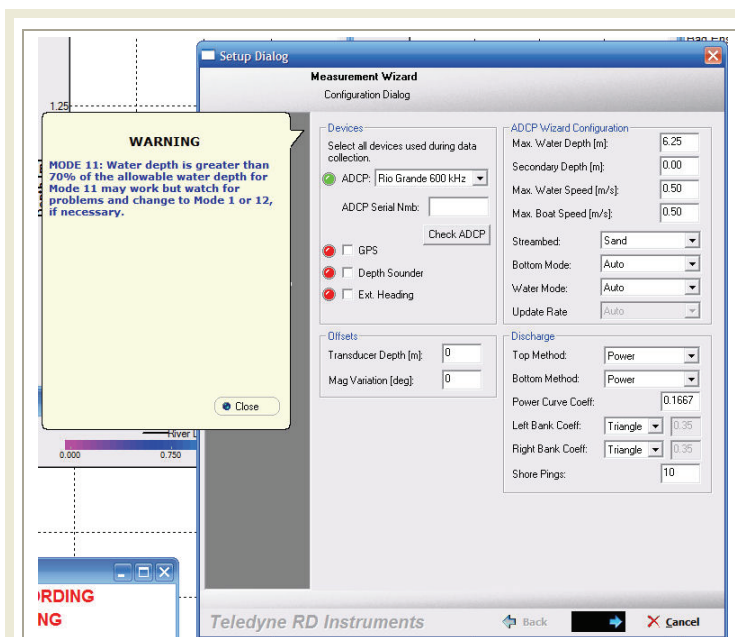
After the Moving Bed Test and the Start/Stop points for the transects have been determined, you may want to make adjustments to the configuration. Perhaps you noticed that the river is deeper than originally entered or you want to use a different Water Mode or Bottom Mode than you used for the moving bed test.



NOTE. You can adjust the **Field Configuration** as long as data has not been collected with the node (it will have **Next Transect XXX**, where XXX is the transect number and the transect check box is grey).



Right-click on **Field Configuration** and select **Configuration Wizard**.



The ADCP communication setting will be checked again and then will open at the **Configuration Dialog** page (see [Configuration Dialog](#)).

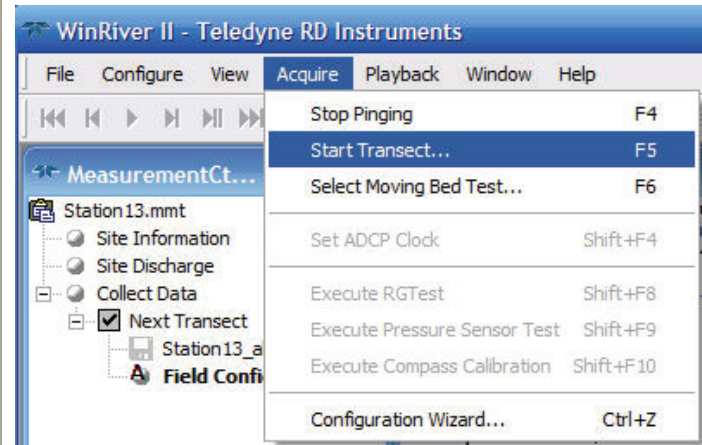
Make changes as needed and click **Next**.

In this example, the **Maximum Water Depth** was increased to 6.25 meters. *WinRiver II* is suggesting that the Water Mode may need to be set to Mode 1 or Mode 12 for best results. If necessary, run another test transect and make further adjustments.

Continue through the wizard by clicking **Next** at each page and then select **Finish** at the Summary page.

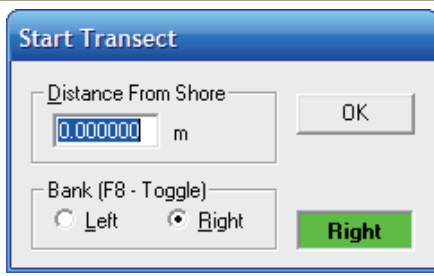
Transects

Best practices encourage a minimum of four high quality transects be collected that agree with each other to within 5% of the mean of all the samples. The following sequence must be repeated for every transect taken over the water body.



Acquire Data – Start Shore

Starting at one of the edge positions determined in [Mark Transect Start and End Points](#), on the **Acquire** menu click **Start Transect (F5)** to begin recording.



When prompted, enter the beginning distance to the bank and determine if this is the left or right bank. When facing downstream, the left bank is on your left side. The edge distance parameters will be saved to the configuration node associated with the raw ADCP data file being recorded.

Click **OK** to start.



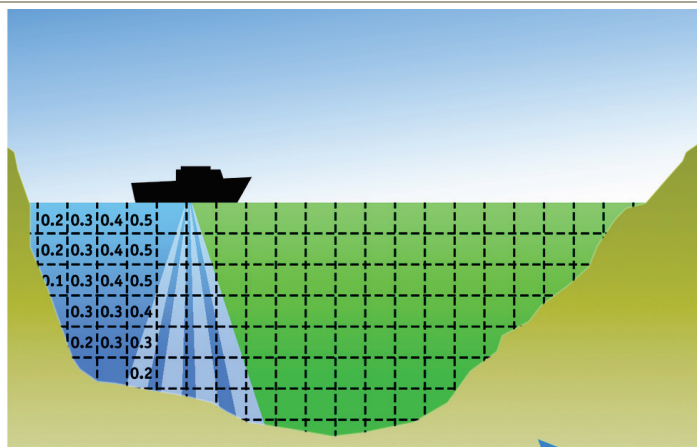
Hold this start position and view the **Composite Tabular** display.

Check if the **Nmb. of Ens.** minus any **Bad Ens.** is more (or equal) to the number of shore ensembles (the default is 10 shore ensembles).



NOTE. If you made a false start, click **F5** to stop recording. Restart at the edge by clicking **F5** to record to a new file. The Left/Right bank toggles at the end of each transect. Click **F8** to toggle the bank to the correct left/right side.

WinRiver II automatically saves the measurement file each time you start/stop a transect. Large numbers of transects in a measurement file may effect computer performance as the measurement file size increases.



Move across the river as smoothly as possible. For the best measurement results, the boat's speed over the bottom should be no greater than the water speed of the river. Pointing the bow of the boat upstream and slowly crabbing across the river will help to maintain a transect path that is perpendicular to the flow.



Move away from the starting point smoothly. If you can not keep the speed as slow and smooth as needed due to a large motor, use a trawling motor instead. Smoothness is most important; On large rivers speed may need to be higher than desired for many reasons.

Composite Tabular 1 - TRDI		
Ens. Nmb.	Nmb. of Ens.	Lost Ens.
422	381	0
Bad Ens.	%Bad Bins	Delta Time
2	0%	0.40
February 24, 2006 12:15:08.59		

Continue across the river until you reach the stop position determined in [Mark Transect Start and End Points](#). Decelerate before reaching the end point such that you do not overshoot it.

You should have discharge values in only the top two depth cells.

Stop at this position and wait for the Shore Ensembles measurements to be recorded.

End Transect

Distance From Shore
 m

Bank (F8 - Toggle)
☒ Left ☐ Right

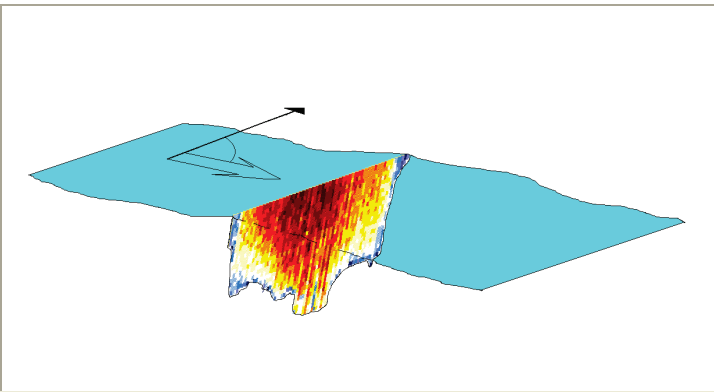
Press **F5** to stop recording.

When prompted, enter the ending distance to the bank. This parameter will be saved to the configuration node.

Composite Tabular 1 - TRDI			
Ens. Nmb.	Nmb. of Ens.	Lost Ens.	
426	385	0	
Bad Ens.	%Bad Bins	Delta Time	
2	0%	0.40	
February 24, 2006 12:15:10.19			

Check **Bad Ensembles** and **% Bad Bins**. The number of Bad Bins should be less than 25%.

The velocity magnitude plot should show a good section with good bottom and velocity data.



Repeat these steps as many times as required for your application. An even number of *at least four* transects are recommended.

When you are finished acquiring the data, press **F4** to stop the ADCP pinging.

Turn off the power to the ADCP and disconnect the cable. Remember to replace the dummy plug to protect the connector.

Step by Step Data Collection

1. Open or create a measurement file.
2. Press **F4** to start pinging.
3. At the start/stop position, press **F5** to start the transect.
4. Enter the starting distance from the shore.
5. Select **Left** or **Right** bank.
6. Wait for 10 shore ensembles.
7. Move across the river.
8. At the stop/start position, wait for 10 shore ensembles.
9. Press **F5** to end the transect.
10. Enter the ending distance from the shore.
11. Repeat steps 3 through 10 to collect at least four transect that agree with each other within 5% of the mean of all the samples.

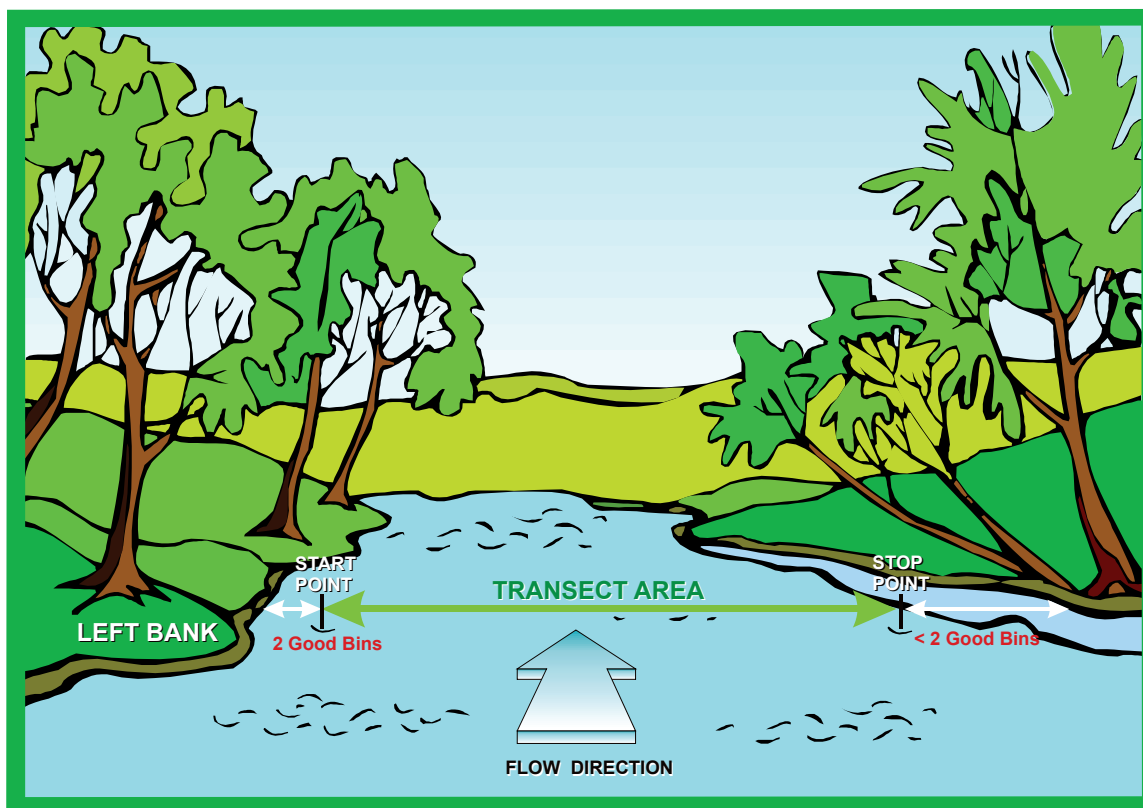


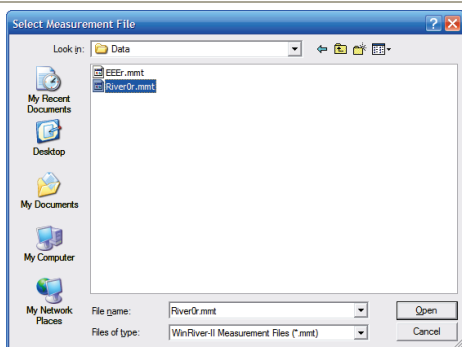
Figure 1. Overview of Data Collection

Data Collections Tips

- Locate the point where a solid two-depth cell measurement can be measured on both banks. Stake or otherwise mark these locations. They represent the starting and stopping points for the transects.
- Accurately measure and enter the **Distance from Shore** when prompted.
- Minimize the ADCP movement while Shore Ensembles are recorded.
- When departing from the edge, slowly accelerate the boat away from the edge and when approaching the other edge slowly reduce the speed such that the boat decelerates and does not overshoot the edge. The goal is to go from edge to edge and not overshoot at either edge. Doing so will allow you to obtain the most accurate measurements, in particular the area measurements.
- Move the ADCP at a slow steady pace in the water during transects.
- Collect a minimum of four transects that agree with each other to within 5% of the mean of all the samples.

Viewing Data with *WinRiver II*

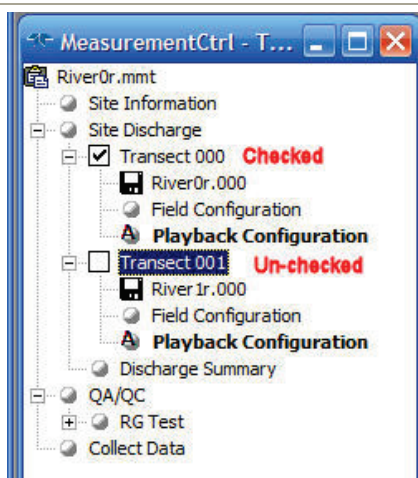
WinRiver II is used for post-processing data to get a total discharge value for the channel. After collecting four transects for each station on the water body, each file must be verified to be within 5% of the mean discharge calculated for the set. If any of the transects are outside of the tolerance, additional transects should be measured.



Start *WinRiver II*.

On the **File** menu click **Open Measurement...**

Select the measurement file (*.mmt) to be played and click **Open**.

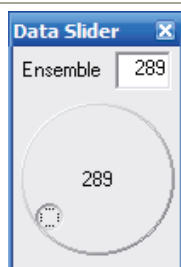


To playback a data file, use the **Playback** menu and select **Reprocess Checked Transects** (click the check box to select the files). This will create a copy of the **Field Configuration node** and creates a **Playback Configuration node**.

The data files will automatically play to the end of the file.

To playback a single data file, click the transect to select it and then use the **Playback** menu and select **Reprocess Selected Transect**.

To playback the next transect file, on the **Playback** menu and select **Reprocess Next Transect**.



On the **Playback** menu, click **First Ensemble** to go to the beginning of the data file. Click **Play** to review the data. To quickly play through the data, on the **Playback** menu, select **Slider** or drag the ensemble marker on the contour plot.

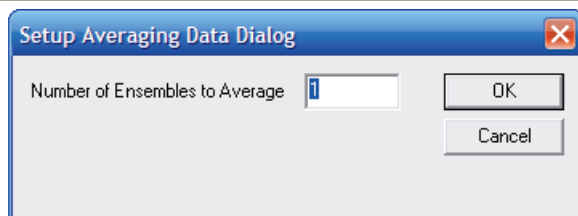


The playback tool bar also has functions to start, stop, rewind, and go to the end of the data file.

Reprocessing Data

WinRiver II is used for post-processing data to get a total discharge value for the channel. Common post-processing tasks include changing your averaging interval, subsection the data to remove bad ensembles or show only a section of the river, export data for use in other programs, screen data, and make corrections to the configuration nodes.

Averaging Data



Right-click on **Site Discharge** and click **Averaging Data**.

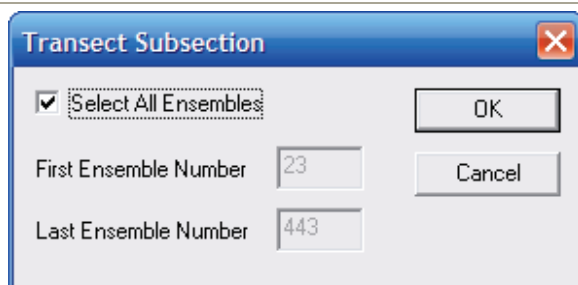
Increase the **Number of Ensembles to Average**.

Click **OK**. Playback/reprocess the data files.



Single ping ensembles are recommended for data collection.

Transect Subsection



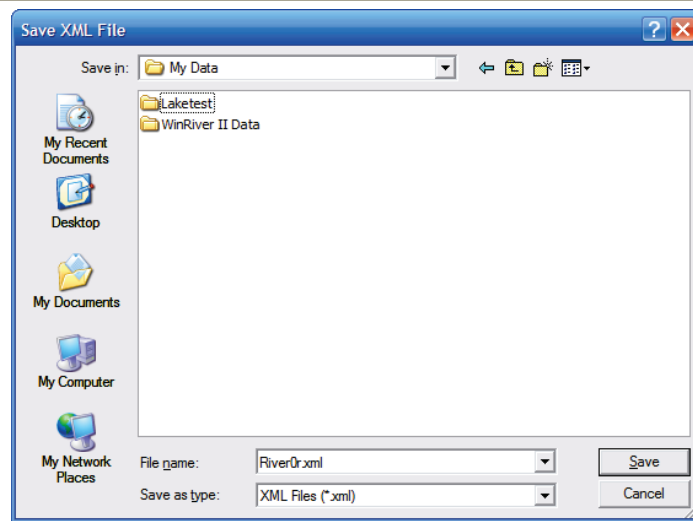
Select the transect file to be subsectioned on the **Measurement Control** window.

Right-click on **Transect** and click **Transect Subsection**.

To select a portion of the data file, uncheck the **Select All Ensembles** box. Enter the **First Ensemble Number** and **Last Ensemble Number** and select **OK**. The file will be reprocessed automatically.

To return to the entire data file, right-click on **Transect** and click **Transect Subsection**. Check the **Select All Ensembles** box. Click **OK**. The file will be reprocessed automatically.

HYDROML Export



Playback/reprocess the data file.

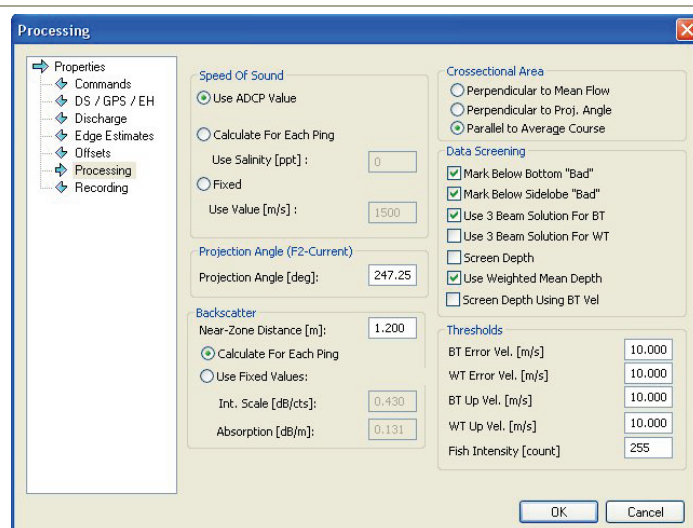
On the **File** menu select **Export as HYDROML**.

Select where to save the file and click **Save**.



HYDROML.XML provides the Hydrologic Scientific Community with a standard structure to allow the definition of hydrologic information.

Data Screening



To change the data screening for a selected data file, right-click on the **Playback Configuration node** and select **Properties**.

Select the **Processing** page. Change the settings as needed.

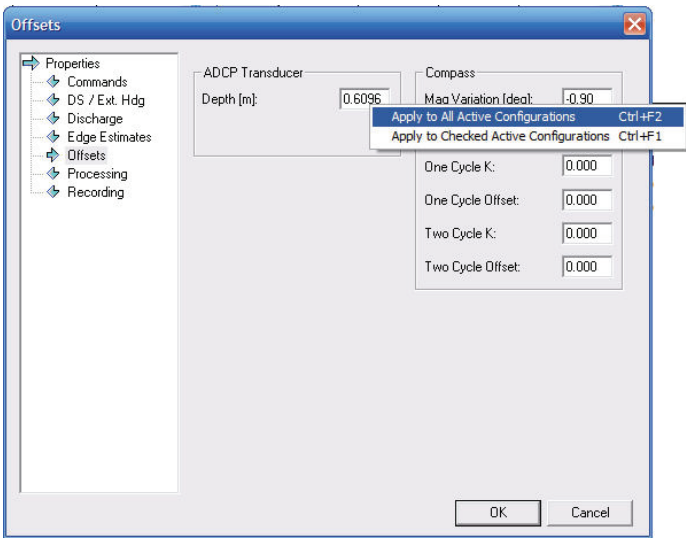
Click **OK**.

The selected data file will playback automatically.



The settings will apply only to the **Playback Configuration node**; the raw data file will not be changed.

Corrections to the Playback Configuration Node



Right-click on the **Playback Configuration node** and select **Properties**.

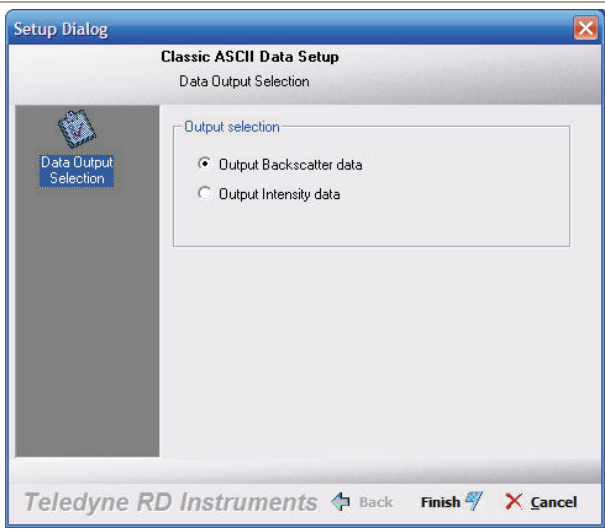
Make the correction.

If the change applies to only one **Playback Configuration node**, click **OK**.

If the change applies to multiple **Playback Configuration nodes**, then right-click the edited item and select **Apply to All Active Configurations** (this makes the correction to all of the configuration nodes: checked or not checked) or **Apply to Checked Active Configurations** (the correction applies only to checked configuration nodes).

Click **OK**.

ASCII-Out



Start *WinRiver II* and load a measurement file.

On the **Configure** menu, click **ASCII Output**, **Classic ASCII Output**.

Select **Output Backscatter** data or **Output Intensity** data.

Click **Finish**.

Playback / reprocess the desired transect.

 The **Generic ASCII Output** allows you to select what ASCII data and in what order you would like it to be displayed in the file. See the *WinRiver II* User's Guide for details.

If you see an error message "*The File does not exist!*" when you double-click the *_ASC.TXT node, this means the file must be played / reprocessed first.

Discharge Summary

Discharge Summary - TRDI							
Transect	Start Bank	# Ens.	Start Time	Total Q m ³ /s	Delta Q %	Top Q m ³ /s	Meas. Q m ³ /s
Riv000	Right	524	12:15:54	1494.734	0.53	164.698	1135.093
Riv001	Left	385	12:12:36	1478.918	-0.53	165.732	1150.510
Average		454		1486.826	0.00	165.215	1142.801
Std Dev.		98		11.184	0.75	0.732	10.902
Std./ Avg.		0.22		0.01	0.00	0.00	0.01

The **Discharge Summary** is an easy way to review transects. It is opened by selecting **View, Discharge Summary** or by using the shortcut key **F12**.

The discharge summary shows all recent transects that have been made and all relevant information on them. It can be used for data collection and playback and is very effective at establishing whether a measurement is good or not.

To add or remove transects from the summary, use the **Measurement Control** window. Checking the **Transect XXX** box (where XXX is the transect number) will add the transect; unchecking the box will remove the transect from the summary.



As a standard for all ADCP users it is widely accepted that transects should be within 5% of each other assuming constant stage.

Using the WinRiver II Q Measurement Summary

The Q Measurement Summary creates a summary of the measurement that can be printed.

Station Number: _____ Meas. No: 0
Station Name: _____ Date: _____

Party: _____	Width: 158.9 m	Processed by: _____
Boat/Motor: _____	Area: 1398.0 m ²	Mean Velocity: 1.07 m/s
Gage Height: 0.000 m	G.H. Change: 0.000 m	Discharge: 1,500 m ³ /s

Area Method: Avg. Course	ADCP Depth: 0.610 m	Index Vel: 0.00 m/s	Rating No.: 1
Nav. Method: Bottom Track	Shore Ens.:10	Adj. Mean Vel: 0.00 m/s	Qm Rating: U
MagVar Method: None (-0.9°)	Bottom Est: Power (0.1667)	Rated Area: 0.000 m ²	Diff.: 0.000%
Depth Sounder: Not Used	Top Est: Power (0.1667)	Control1: Unspecified	Control2: Unspecified
		Control3: Unspecified	

Screening Thresholds:	ADCP:
BT 3-Beam Solution: YES	Type/Freq.: Rio Grande/600 kHz
WT 3-Beam Solution: NO	Serial #: _____
BT Error Vel.: 0.10 m/s	Bin Size: 50 cm
WT Error Vel.: 1.07 m/s	Blank: 25 cm
BT Up Vel.: 0.30 m/s	BT Mode: 5
WT Up Vel.: 3.66 m/s	WT Mode: 1
Use Weighted Mean Depth: YES	WT Pings: 1
	WV: 281

Performed Diag. Test: YES
Performed Moving Bed Test: NO
Performed Compass Test: NO
Meas. Location: _____

Project Name: River0r.mnt
Software: 1.00.12.03

Tr #	Edge Distance		#Ers	Discharge						Width	Area	Time		Mean Vel.		% Bad	
	L	R		Top	Middle	Bottom	Left	Right	Total			Start	End	Boat	Water	Ens	Bins
000 R	30.5	15.2	524	165	1135	111	73.4	11.1	1495	158.9	1398.0	12:15	12:19	0.57	1.07	0	1
Mean	30.5	15.2	524	165	1135	111	73.4	11.1	1495	158.9	1398.0	Total	00:03	0.57	1.07	0	1
SDev																	
R/M%																	

Remarks: _____

Start *WinRiver II* and open a measurement file.

To add or remove transects from the summary, use the Measurement Control widow. Checking the **Transect XXX** box will add the transect; un-checking the box will remove the transect from the summary.

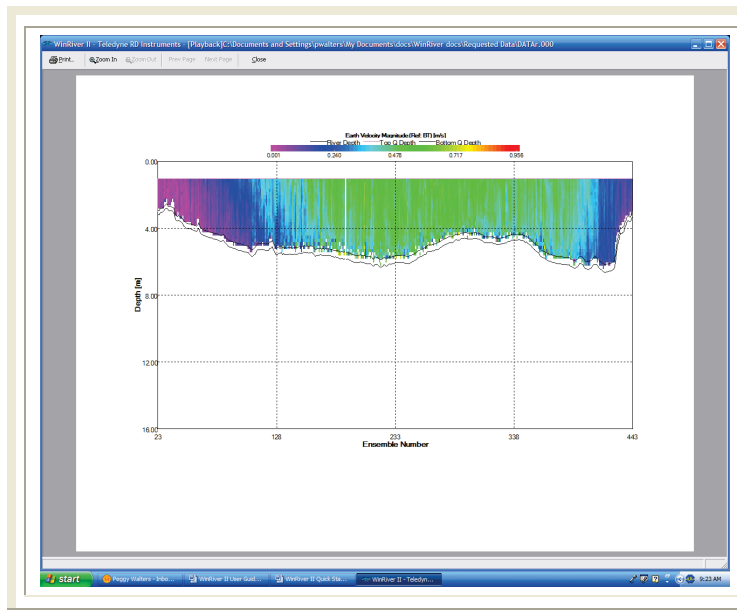
Playback / Reprocess the transects; use the **Playback** menu and select **Reprocess Checked Transects**.

Check/edit the Site Information by right-clicking on **Site Information** in the Measurement Control window and selecting **Site Wizard**.

On the **File** menu, click **Print Preview Q Measurement Summary**.

On the **Q Measurements** screen, click **Print** to print a copy of the discharge summary.

Print a Plot or Display



Click on the plot/display to be printed. The title bar will be highlighted.

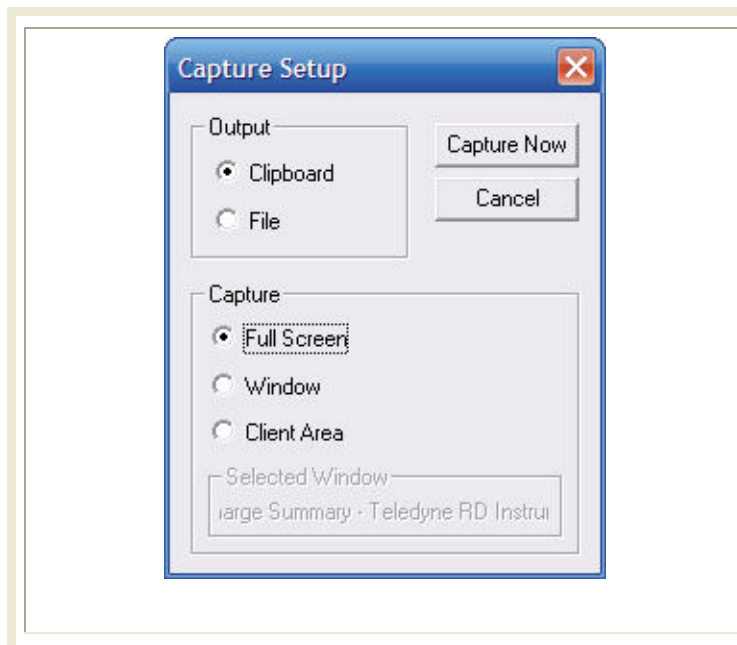
On the **File** menu, click **Print Setup**. Select the desired setting and printer.

On the **File** menu click **Print Preview**. If the display is acceptable, click **Print**.



Choose **Landscape** for contour plots.

Screen Captures



If you would like to save a plot/display for use in other documents, do the following.

Click on the plot/display to be saved. The title bar will be highlighted.

On the **Configure** menu, select **Screen Capture**. This will bring up the **Capture Setup** dialog box.

Select **Clipboard** or **File**.

Select **Full Screen**, **Window**, or **Client Area**. The **Client Area** will include all parts of the graph/plot except the title bars.

Click **Capture Now**. If you selected **File**, name the file and click **Save**.

Click **Cancel** to exit this dialog box.

Water Profiling Modes

This section explains all of the water-profiling modes available for the WorkHorse Rio Grande, Mariner ADCPs, and WorkHorse ADCPs with the high-resolution water profiling upgrade installed. For each mode, we provide a general description, an explanation of the best place to use this mode, specifics about the mode, and any setup considerations. Use Table 1 as a guide for choosing the appropriate mode for your water flow conditions.

Typically if the flow is slow and the depth is shallow you would first try Mode 11. If the flow were too fast or turbulent for Mode 11 you would use Mode 12 if fitted. If Mode 11 or Mode 12 is not suitable then Mode 1 will work in all but the most extreme situations. Mode 5 and Mode 8 are still included for backward compatibility and for users who are familiar and satisfied with their performance.

Table 1: Recommended River Water Profiling Modes

	Mode 1	Mode 12	Mode 11	Mode 13	Mode 5	Mode 8
Typical application	Fast water of all depths. Rough and dynamic situations. Good in streams too fast or deep for modes 5, 8 & 11 or where Mode 12 has problems.	Fast water of all depths. Good in streams too fast or deep for modes 5, 8 & 11. Good for deep, slow water. See Note 1.	Slow, shallow streams with velocities < 1.0 m/sec (depth dependant) with low shear and/or turbulence.	Slow, shallow streams where the Maximum Stream Depth is ≤ 1.0 meters and the Maximum Stream Velocity is < 0.25 m/s	Slow, shallow streams with velocities < 0.5 m/sec with low shear and/or turbulence.	Shallow streams with velocities < 1 m/sec and with moderate shear (rough bed) and/or turbulence.
Minimum recommended cell size (meters)	0.50* 0.25	0.25* 0.10	0.10 0.05	StreamPro ADCPs Only See Low Noise Mode/Water Mode 13	0.10 0.10	0.10 0.10
Recommended Cell Size (meters)	0.50 0.25	0.25 0.10	0.25 0.05		0.25 0.10	0.25 0.10
Single ping standard deviation (cm/s) (using rec. cell size)	13.62 13.64	6.24 6.95	0.74 1.34		0.33 0.44	3.34 5.15
First range cell (meters)	0.97 0.51	0.73 0.26	0.49 0.09		0.49 0.14	0.49 0.14
Minimum profiling range (meters) Bottom Mode 5	1.7 1.0	1.7 1.0	1.6 0.9		1.6 0.9	0.9 0.6
Minimum profiling range (meters) Bottom Mode 7	NA 0.7	NA 0.5	NA 0.3		NA 0.7	NA 0.3
Maximum profiling range (meters)	73.1 19.55	68.29 15.82	<8.0 <4.0		<8.0 <4.0	<8.0 <4.0
Maximum relative velocity (m/s)	10 m/sec	10 m/sec	1 m/sec (Depth Dependent)		0.5 m/sec	1 m/sec

* 600 kHz values are in **bold** font, and 1200 kHz values are in regular font.

Specifications are for 25 cm blank (600 kHz), 5cm Blank (1200 kHz), 10° C temperature, and 0.0 salinity.

Note 1. Mode 12 table assumes 20 sub-pings (WO20,4).



NOTE. The ranges in Table 1 are measured from the transducer face. Add the transducer depth to determine the actual minimum and maximum profiling depths.



NOTE. Maximum range depends on water temperature and depth cell size. Use *PlanADCP* to compute the maximum range for a particular ADCP set-up and water temperature. The standard deviation of modes 5, 8 and 11 varies with water speed, boat speed, bed-form roughness, channel depth, and turbulence.

Commonly Used ADCP Commands

The commands shown in Table 2 (these are only a small percentage of those available) can all be found with detailed descriptions in the WorkHorse Commands and Output Data Format guide, however below are some of the more commonly used commands.

Table 2: Commonly Used ADCP Commands

Command	Description
BX	Max depth in decimeters (meters x 10) that the ADCP will look for the bottom - set this greater than your deepest depth.
WS	Depth cell size (cm)
WN	Number of depth cells (WS x WN = BX x 10)
WF	Blanking distance (cm)
WV	Ambiguity velocity (160 to 200 is usually good) (approximately water velocity in cm/sec)
WM	Water mode 1, 5, 8, 11 or 12 (1200 Rio Grande / Zed Head)
WP	Water pings
BP	Bottom pings
CL	Sleep Between Pings (CL0 = Do Not Sleep, CL1 = Sleep Between Pings)

Commonly Used BBTalk Commands

Once *BBTalk* is running and connected to the ADCP, commands can be entered to interrogate the unit. The commands shown in Table 3 (these are only a small percentage of those available) can all be found with detailed descriptions in the WorkHorse Commands and Output Data Format guide.

Table 3: Commonly Used BBTalk Commands

Command	Description
CR1	Parameters set to FACTORY Settings
CK	Parameters saved as USER defaults
PS0	Basic Instrument Information including Serial Number, Frequency, Firmware, and Board Serial Numbers
PT200	Built in Self Test
CB	Change baud rate
PC1	Beam Continuity Test - Rub beams when asked to do so. ADCP must be in air to run this test.
PC2	Position, direction, temperature etc - Press any key to quit
PA	Pre-deployment Tests
?	List possible inputs. If added at the end of a command (i.e. CB?), it will list the command setting.
CS	Start Pinging. Once ADCP is pinging, send a Break to stop.
= = =	Send Break (Radio Modem)
End	Send Break (Direct Cable)

WinRiver II Shortcut Keys

Key	Description
F1	Help
F2	Reprocess transect
F3	Configuration Setting
F4	Start/Stop Pinging
F5	Start/Stop Transect
F6	Moving Bed Test
F7	Properties
F8	Toggle Bank
F9	Toggle Ensemble Header Tabular view
F11	Toggle Detailed Discharge/Composite Tabular view
F12	Toggle Discharge Summary Tabular view
Ctrl-A	Output ASCII data file
Ctrl-A	Set as Active Configuration (when Measurement Control window selected)
Ctrl-B	Reference - Bottom Track
Ctrl-C	Copy
Ctrl-D	Toggle Acquire Control window
Ctrl-E	Close Measurement File
Ctrl-F	Create Measurement from Data Files
Ctrl-F1	Apply to Checked Active Configurations
Ctrl-F2	Apply to All Active Configurations
Ctrl-F11	Select Stick (when Ship Track plot selected)
Ctrl-F11	Add Transect (Playback) (when Measurement Control window selected)
Ctrl-F11	Add Contour Pane (when Contour plot selected)
Ctrl-F12	Remove Contour Pane (when Contour plot selected)
Ctrl-F3	Toggle Measurement Management window
Ctrl-F5	Reprocess Checked Transects
Ctrl-F7	Properties (when window or file selected)
Ctrl-F8	Data Selection (when Profile or Contour plot selected)
Ctrl-F8	Transect Subsection (Playback) (when Measurement Control window selected)
Ctrl-F9	Averaging Data
Ctrl-G	Reference - GPS (GGA)
Ctrl-H	Export as HYDROML
Ctrl-U	Duplicate Configuration node (when Measurement Control window selected)
Ctrl-T	Delete Configuration node (when Measurement Control window selected)
Ctrl-J	Rename Configuration node (when Measurement Control window selected)
Ctrl-R	Reset Configuration node Properties (when Measurement Control window selected)
Ctrl-I	Site Wizard
Ctrl-K	Add Note (when Measurement Control window selected)
Ctrl-L	Lock/Unlock Measurement File
Ctrl-M	View Command Log
Ctrl-N	Reference - None
Ctrl-O	Open measurement file
Ctrl-P	Print
Ctrl-PgDn	Scale Sticks Down (when Ship Track plot selected)
Ctrl-PgUp	Scale Sticks Up (when Ship Track plot selected)
Ctrl-S	Save Measurement File
Ctrl-V	Reference - GPS (VTG)
Ctrl-W	New Measurement
Ctrl-Z	Configuration Wizard
Shift-F4	Set ADCP Clock
Shift-F5	Reprocess Selected Transect
Shift-F6	Reprocess Next Transect
Shift-F8	Execute RGTest
Shift-F9	Execute Pressure Sensor Test
Shift-F10	Execute Compass Calibration
Minus	Previous Ensemble (Playback)
Space	Next Ensemble (Playback)
Ctrl-Space	Several Ensembles (Playback)
Ctrl-Home	Slider / Go to Ensemble (Playback)
Home	First Ensemble (Playback)
End	Last Ensemble (Playback)

Where to Find More Information

Congratulations! You have completed the Quick Start Guide. For more detailed information, see the following sections in the *WinRiver II* User's Guide.

Using the Measurement Wizard – This section covers how to create a configuration node using the **Measurement Wizard** and a detailed explanation of the configuration node settings.

Acquiring Discharge Data – This section is a more detailed explanation of the steps covered in the Quick Start Guide.

Post-Processing of Discharge Data – This section covers data playback, processing, and screen captures.

Integrating Depth Sounder, External Heading, and GPS Data – Use this section to integrate External heading, GPS data, and depth sounder data into your real-time discharge calculations. These devices are used when environmental conditions make it difficult to get unbiased boat velocity and/or depth using bottom tracking.

ADCP Commands – Use this section for a description of each command and guidelines for setting these commands to acquire reliable discharge data.

Water Profiling Modes – This section explains all of the water-profiling modes available for the WorkHorse Rio Grande, Mariner ADCPs, and WorkHorse ADCPs with the high-resolution water profiling upgrade installed. For each mode, we provide a general description, an explanation of the best place to use this mode, specifics about the mode, and any setup considerations.

NOTES